Adrenal physiology

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Modified by Hugh Clements-Jewery PhD based on the original by Carlos Stocco, PhD

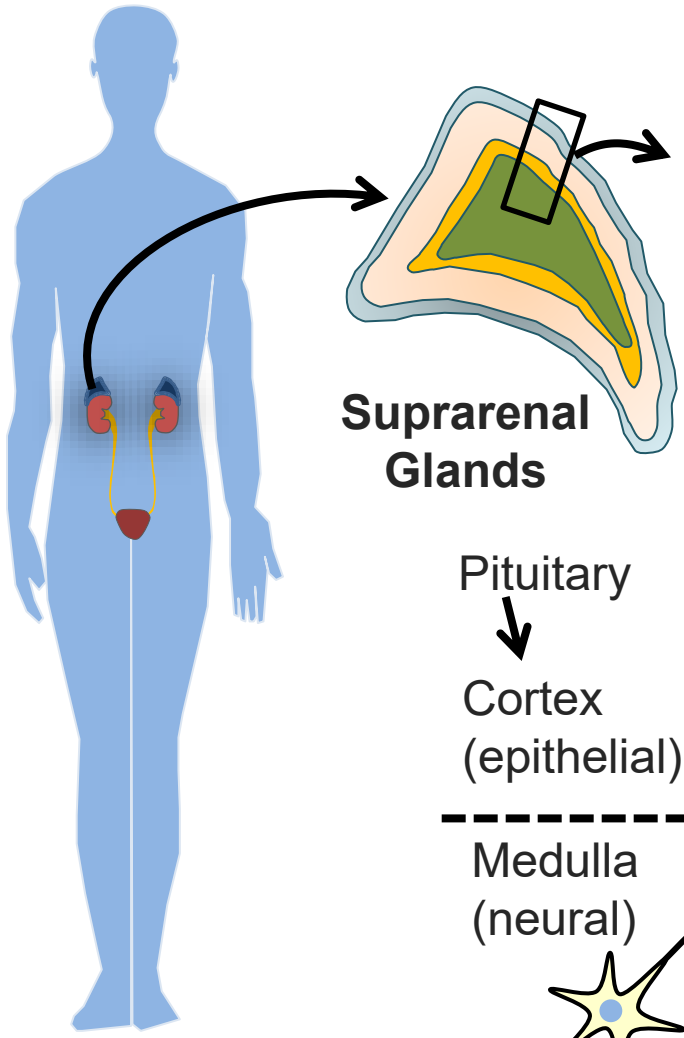
Objectives

1. Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.
2. Describe the cellular mechanism of action of adrenal cortical hormones.
3. Describe the regulation of cortisol secretion including how exogenous steroid drugs affect adrenal cortical function.
4. Describe the target tissues and physiologic actions of cortisol.
5. Describe the regulation of aldosterone release.
6. Describe the effect of aldosterone on sodium and potassium balance and the physiological stimuli that modulate its secretion.
7. Describe the physiologic role of adrenal androgens.
8. Describe the effect of ACTH on adrenal androgen production.
9. Compare and contrast Congenital Adrenal Hyperplasia (CAH) caused by deficiency of: 21α-hydroxylase, 11β-hydroxylase & 17α-hydroxylase.
10. Describe the chemical nature of catecholamines, their biosynthesis, and stimulus for secretion.
11. Describe the influence of cortisol on the adrenal medulla and explain its physiological significance during stress.

Key questions to answer from this material:

1. What adrenal hormones are manufactured in which layers of the adrenal cortex?
2. What key enzymes are needed in each layer?
3. What key enzymes are missing in the zona glomerulosa and zona reticularis? Why is this important?
4. How is steroid hormone synthesis increased acutely? How is it increased chronically?
5. What mechanism prevents cortisol from causing hypertension?
6. What conditions lead to congenital adrenal hyperplasia and why? What effects on salt retention and adrenal androgens are associated with these conditions?
7. What are the key enzymes involved in production of epinephrine in the adrenal medulla? Why is it important that blood flows from cortex to medulla?

The adrenal gland and associated hormones



Suprarenal Glands

Pituitary

Cortex (epithelial)

SALT

Zona Glomerulosa

Mineralocorticoid

Aldosterone

SUGAR

Zona Fasciculata

Glucocorticoids

Cortisol

SEX

Zona Reticularis

Androgens

DHEAS

Medulla (neural)

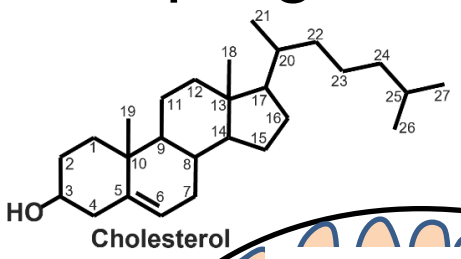
Chromaffin Cells → Epinephrine Norepinephrine

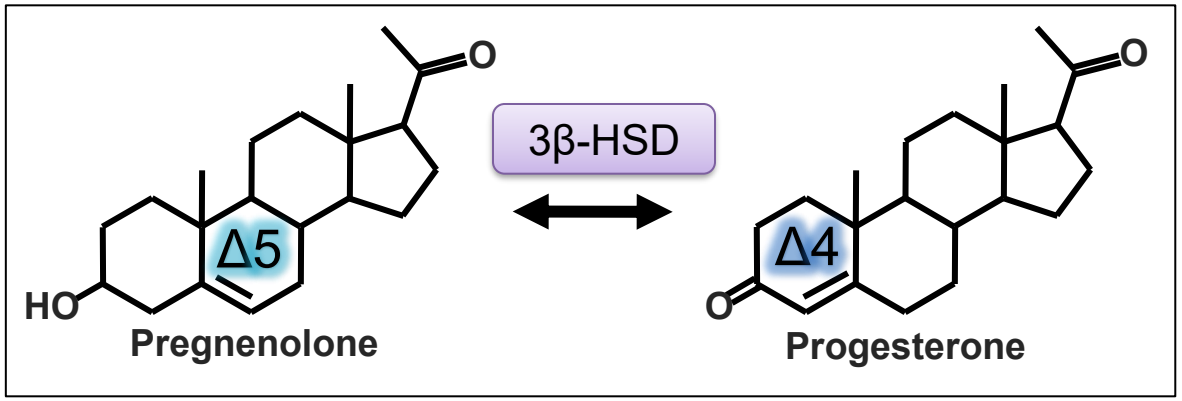
Cholinergic Preganglionic Sympathetic Neurons

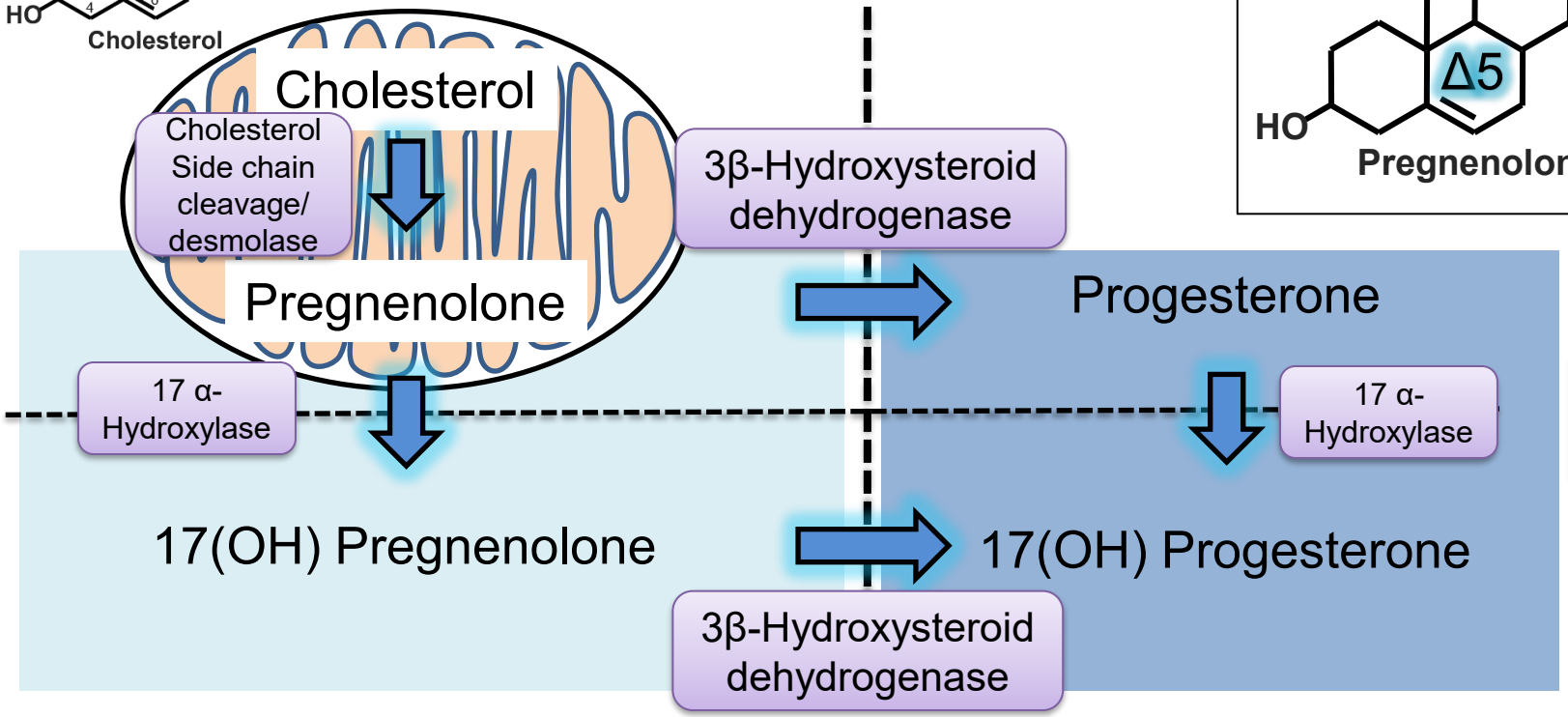
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Overview of the biosynthetic pathways involved in aldosterone, cortisol, and adrenal androgen production

Common steroid precursor pathways: Cholesterol is converted into pregnenolone, and there are two biosynthetic pathways from there





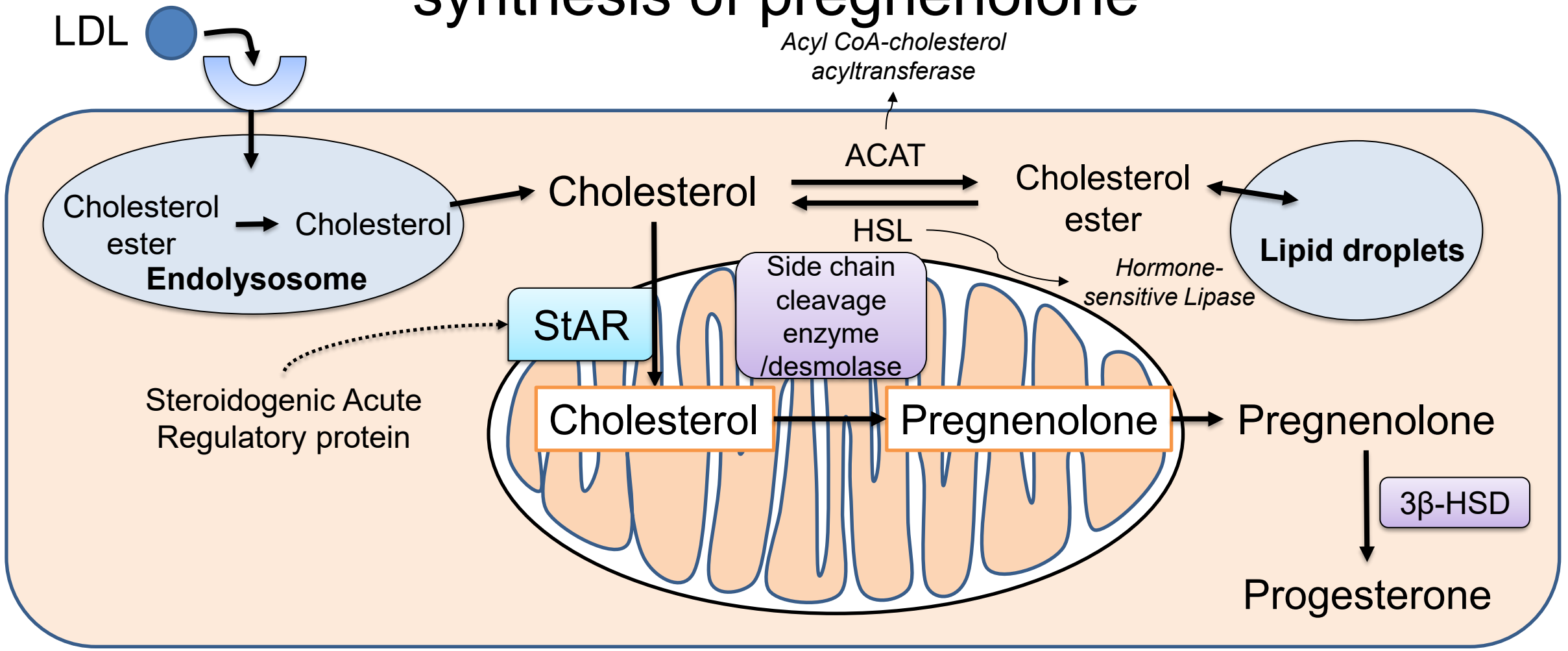


Δ4 pathway
Double bond is in 4-5 position

Δ5 pathway
Double bond is in 5-6 position

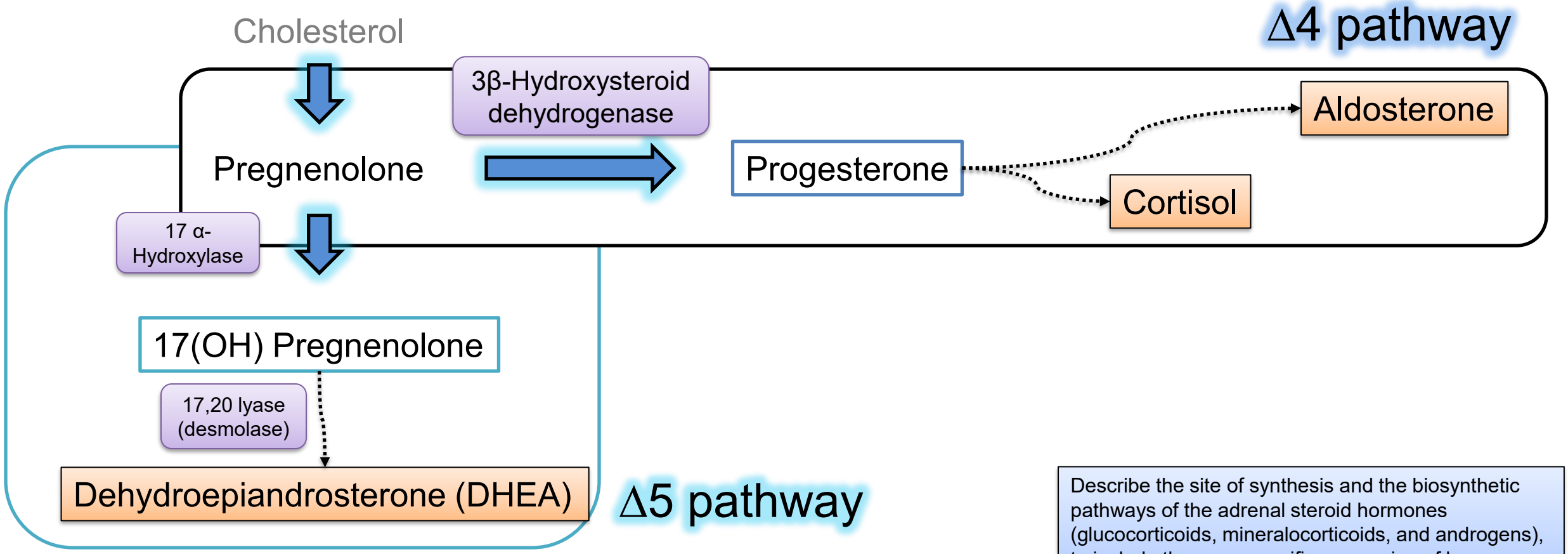
Key concept:
3β-HSD is necessary to convert steroid precursors to Δ4 structures

Cholesterol can be mobilized from intracellular lipid droplets and shuttled into the mitochondria via StAR protein for synthesis of pregnenolone



Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

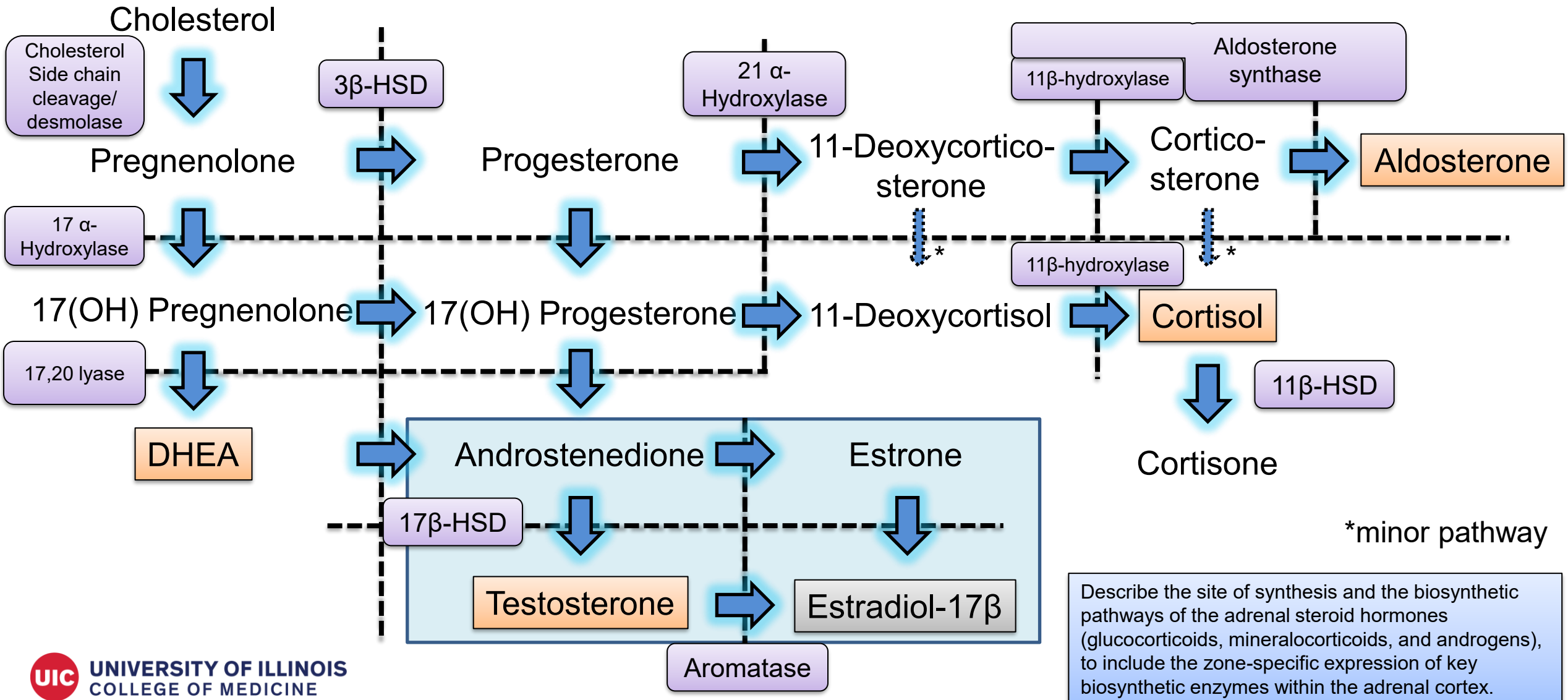
Aldosterone, cortisol and DHEA are synthesized from these precursors



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

Steroid synthesis cheat sheet



*minor pathway

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Enzymes in steroid hormone synthesis can be referred to using a P450 synonym or by gene

Learn these names for Step & the exam!



Enzyme	Synonym	Gene
Cholesterol side-chain cleavage/desmolase	P450scc	CYP11A1
11β-hydroxylase	P450c11	CYP11B1
17α-hydroxylase	P450c17	CYP17
17,20-lyase	P450c17	CYP17
21α-hydroxylase	P450c21	CYP21A2
Aldosterone synthase*	P450aldo	CYP11B2*
Aromatase	P450arom	CYP19

*can also perform 11β-hydroxylase reactions

Q1: Which of the following is an intermediate in the Δ4 pathway, but not the Δ5 pathway?

A. Cholesterol
B. Dehydroepiandrosterone
C. Pregnenolone
D. 17(OH) pregnenolone
E. Progesterone

Blank slide inserted intentionally

Q1 Answer: Which of the following is an intermediate in the Δ4 pathway, but not the Δ5 pathway?

A. Cholesterol
B. Dehydroepiandrosterone
C. Pregnenolone
D. 17(OH) pregnenolone
E. Progesterone

Progesterone is formed from the action of 3β-hydroxysteroid dehydrogenase on pregnenolone. Progesterone is a Δ4 structure. Cholesterol and pregnenolone are common to both the Δ4 and Δ5 pathways. 17(OH) pregnenolone and dehydroepiandrosterone are part of the Δ5 pathway.

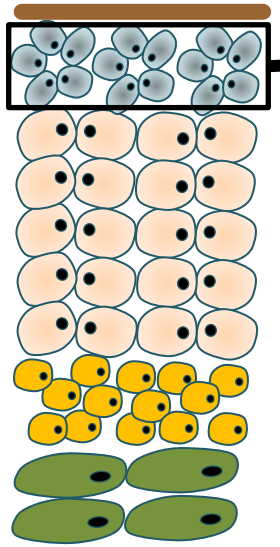
Summary: biosynthetic pathways in the adrenal cortex

1. The adrenal gland consists of 3 cortical layers and a medulla. The zona glomerulosa is the outermost cortical layer and produces aldosterone. The zona fasciculata is the middle cortical layer and produces cortisol. The zona reticularis is the innermost cortical layer and produces adrenal androgens (DHEA).
2. Aldosterone, cortisol and DHEA are manufactured from common steroid precursors – cholesterol and pregnenolone.
3. DHEA is produced by the $\Delta 5$ pathway, whereas aldosterone and cortisol are manufactured using the $\Delta 4$ pathway that requires the enzyme 3β-hydroxysteroid dehydrogenase.
4. Each enzyme has a synonym and gene product associated with it – enzymes can be referred to by any of these names.

Mineralocorticoids

Zona glomerulosa

Effects, site of action and regulation of Aldosterone

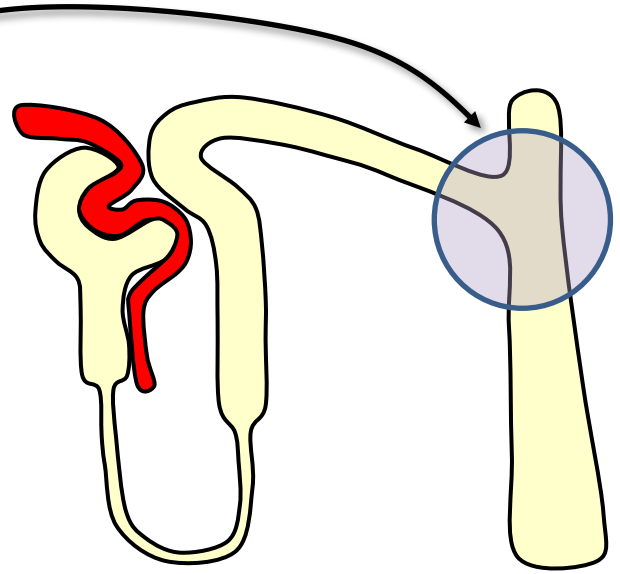


Aldosterone

Major regulators:

- \+ Angiotensin II
- \+ Plasma K^+
- − ANP

NOT regulated by the pituitary



Late distal tubule, connecting segment, cortical collecting duct

Effects:

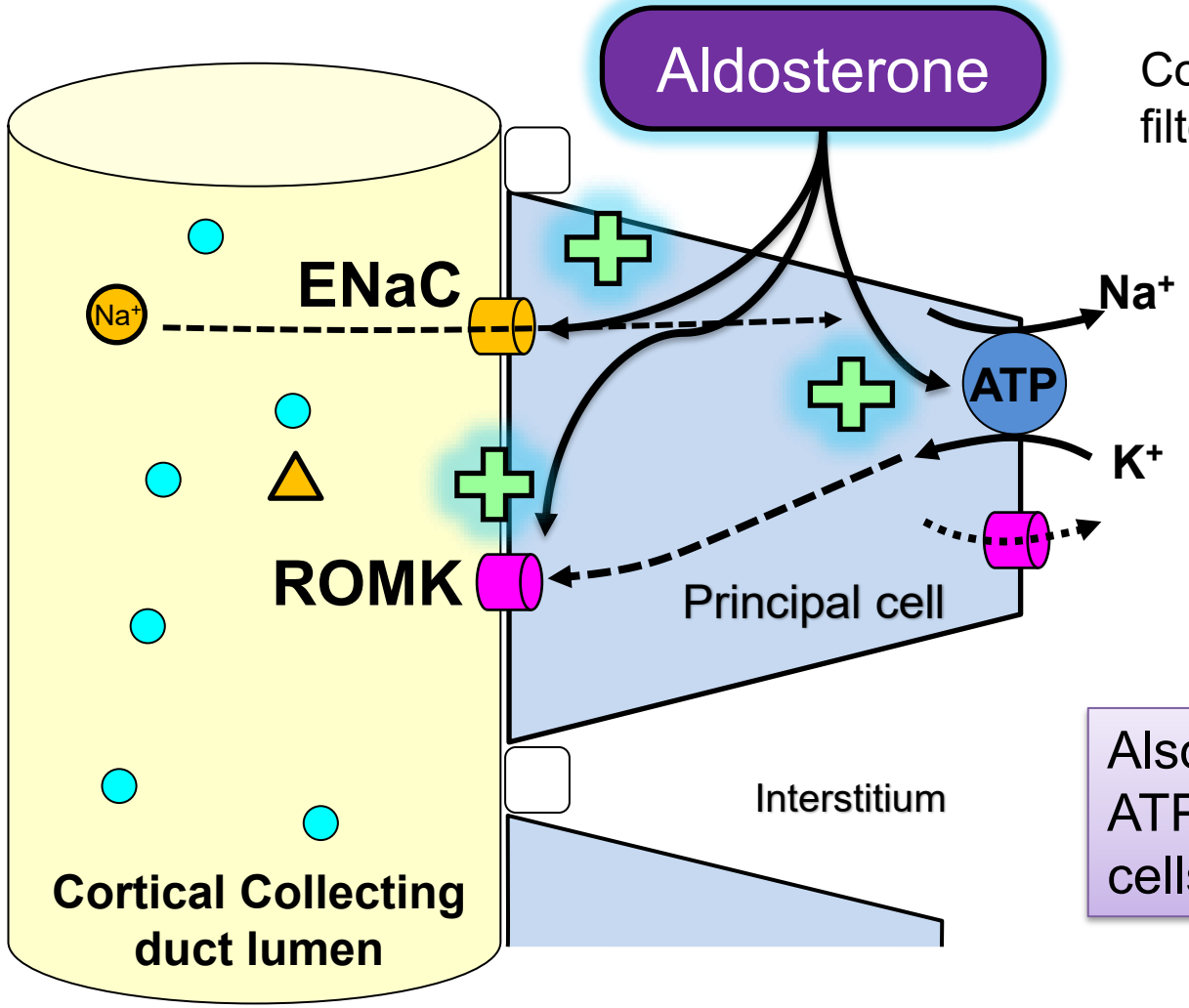
- Na^+ retention
- Regulation of extracellular volume
- K^+ secretion
- Control of potassium homeostasis – maintains excitability of nerves and muscles.

Describe the effect of aldosterone on sodium and potassium balance and the physiological stimuli that modulate its secretion.

Aldosterone stimulates Na^+ reabsorption and K^+ secretion in the Principal cells of the distal nephron

Aldosterone increases:

1. Expression of Na^+/K^+ ATPases
2. Expression of ENaCs
3. Activity of ENaCs
4. Apical K^+ Permeability (ROMK channels)



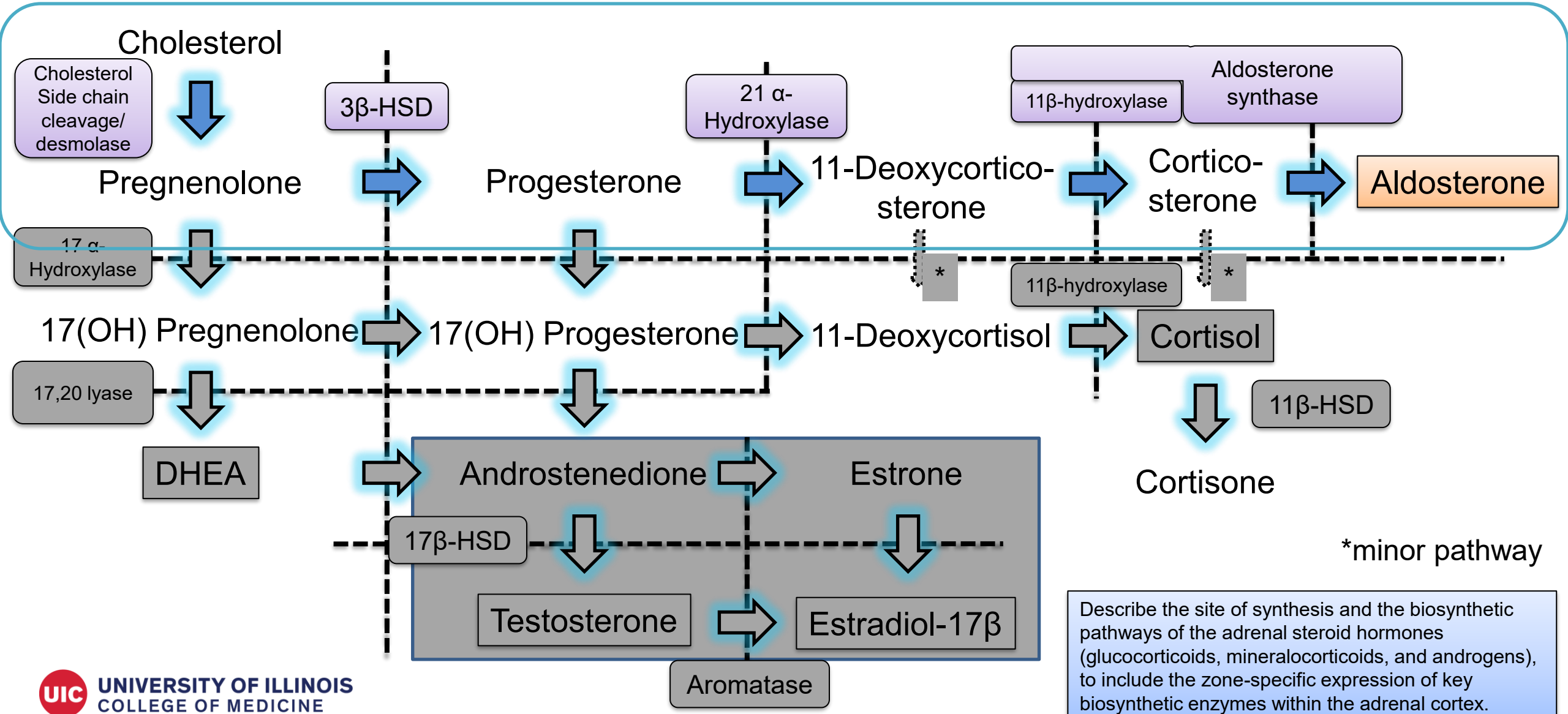
Controls reabsorption of ~2% filtered sodium

Describe the effect of aldosterone on sodium and potassium balance and the physiological stimuli that modulate its secretion.

Also stimulates H^+/K^+ ATPase in intercalated cells (contraction alkalosis)

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Steroid synthesis cheat sheet

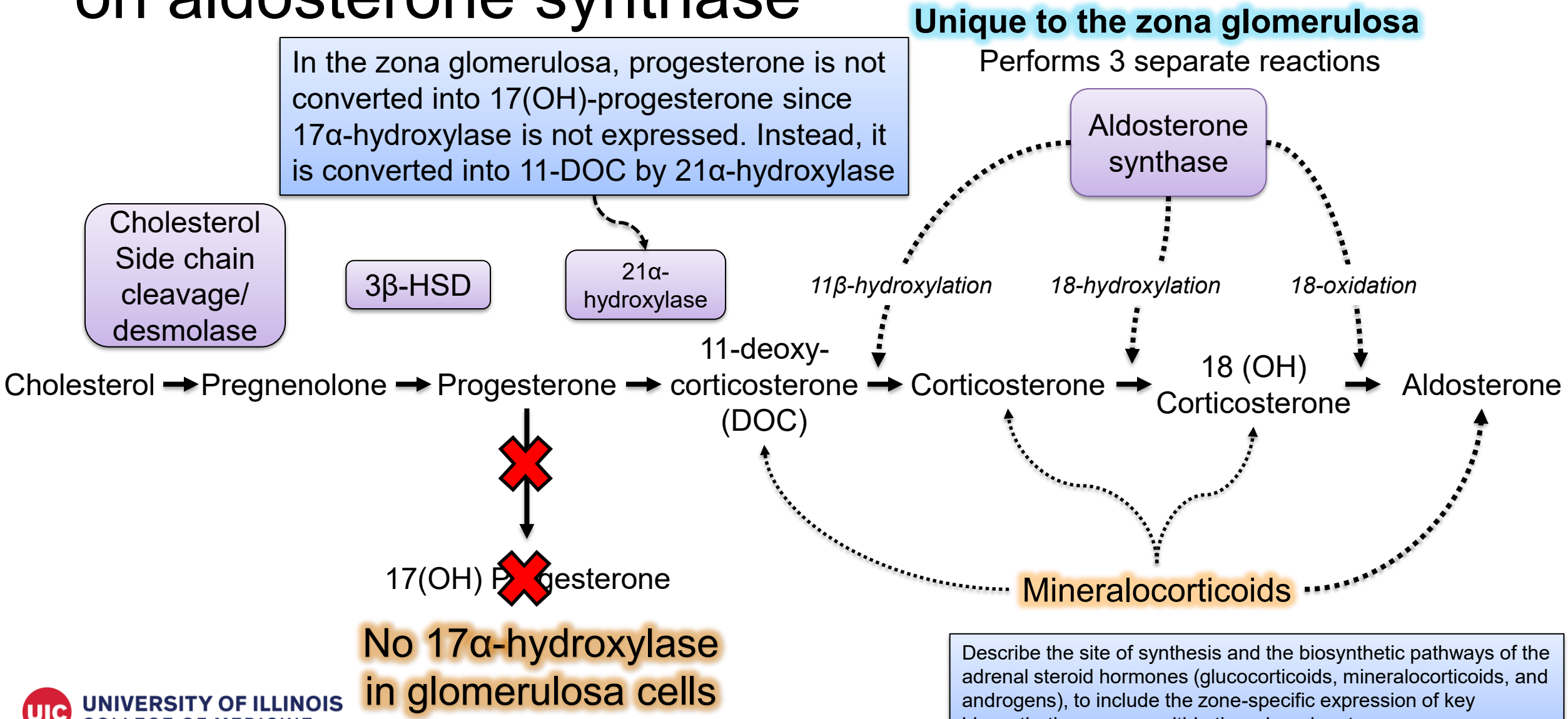


*minor pathway

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

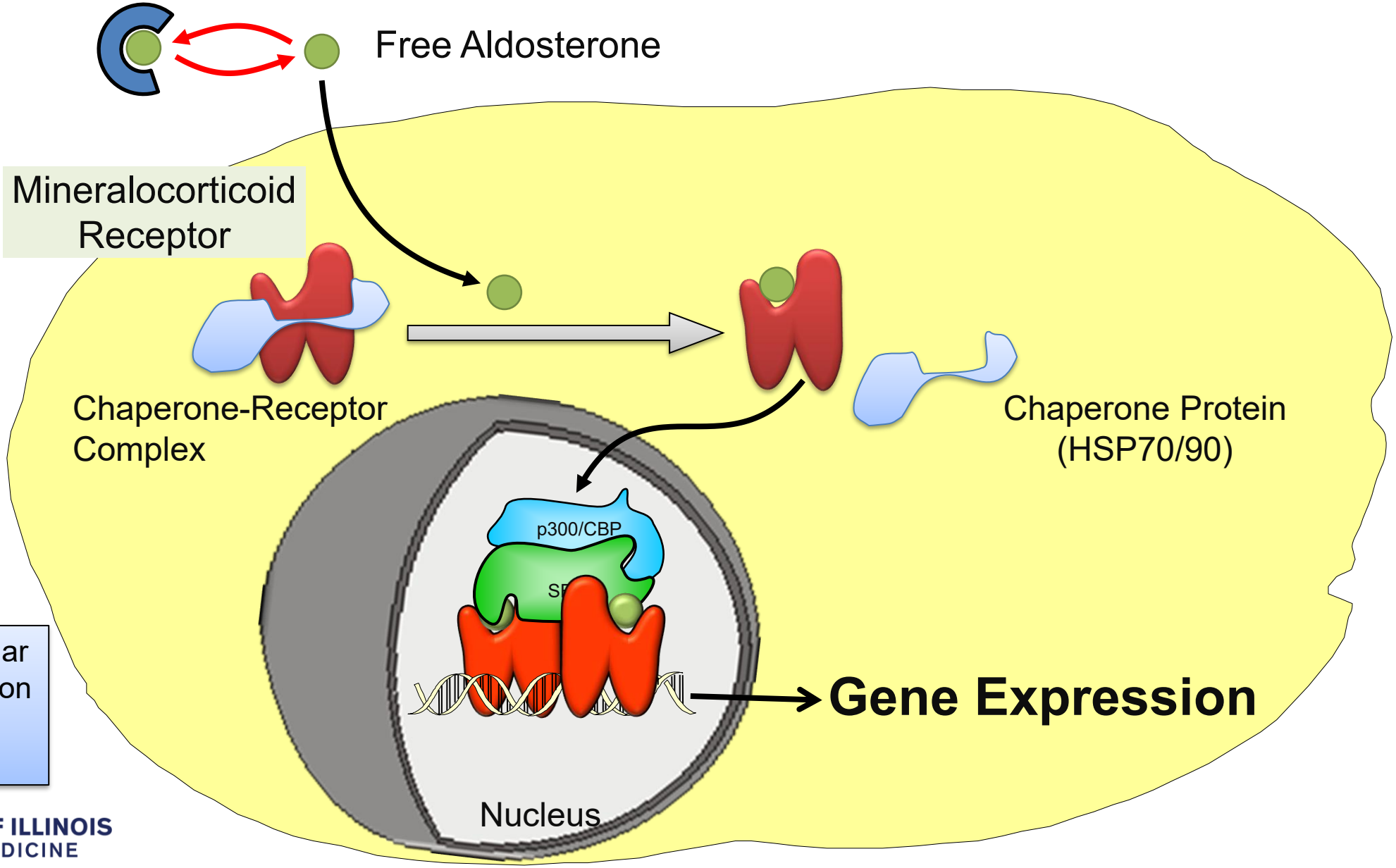
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Biosynthesis of aldosterone depends critically on aldosterone synthase



Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

Mechanism of Aldosterone Hormone Action

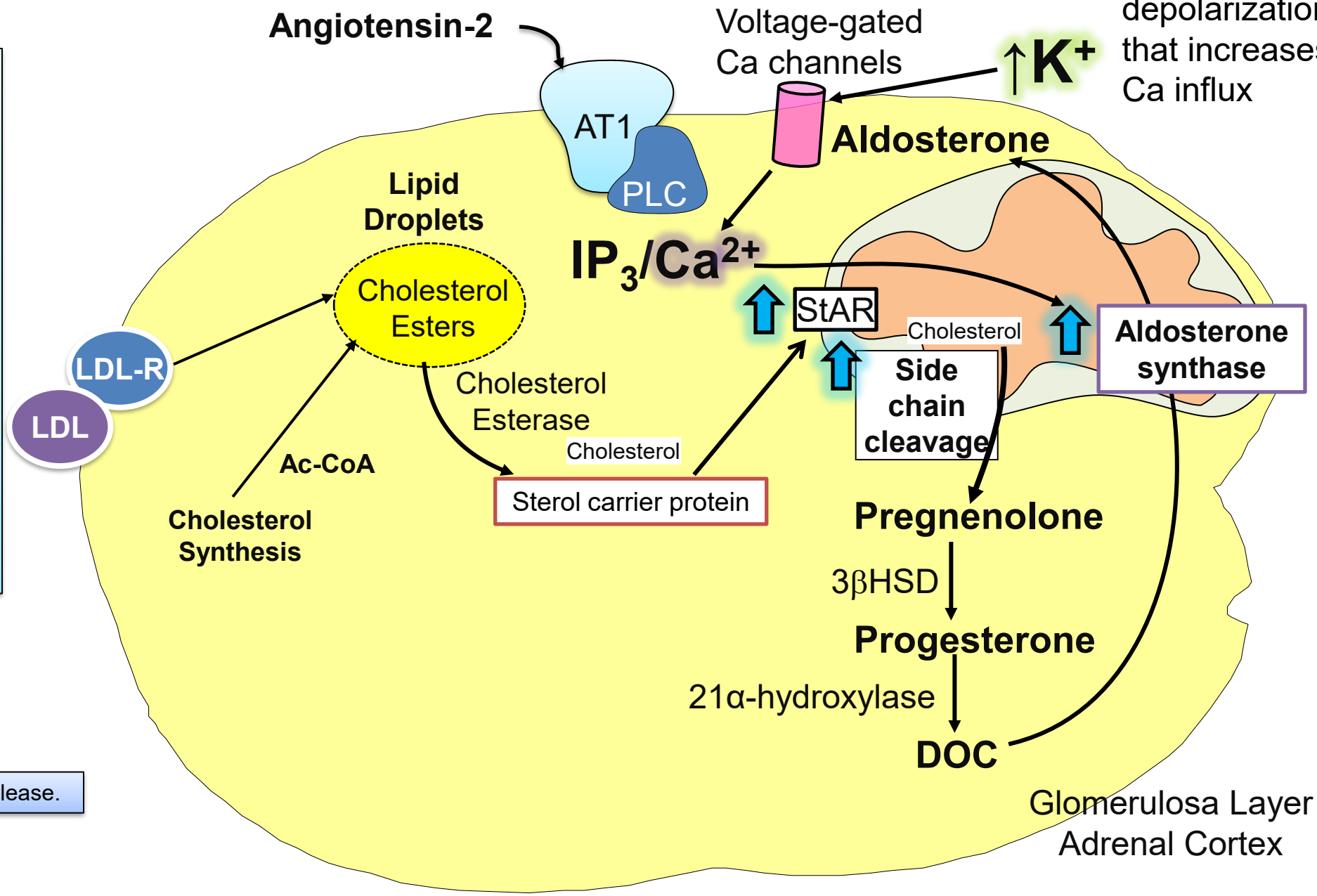


Describe the cellular mechanism of action of adrenal cortical hormones.

Regulation of Aldosterone Synthesis

Ang II stimulation of glomerulosa cells:

1. Increase in Ca^{2+} via IP_3 & opening of voltage-sensitive Ca^{2+} channels
2. Sustained increase in Ca^{2+} increases shuttling of cholesterol into the mitochondria, stimulates side chain cleavage, & increases activity of aldosterone synthase



Describe the regulation of aldosterone release.

Q2: Which of the following intermediates has mineralocorticoid activity?

A. Pregnenolone

B. Progesterone

C. 17(OH) progesterone

D. 11-deoxycorticosterone (DOC)

Q2 Answer: Which of the following intermediates has mineralocorticoid activity?

A. Pregnenolone

B. Progesterone

C. 17(OH) progesterone

D. 11-deoxycorticosterone (DOC)

11-deoxycorticosterone, corticosterone and aldosterone all have mineralocorticoid activity, meaning they have similar actions to aldosterone in terms of Na and K balance.

Summary of zona glomerulosa

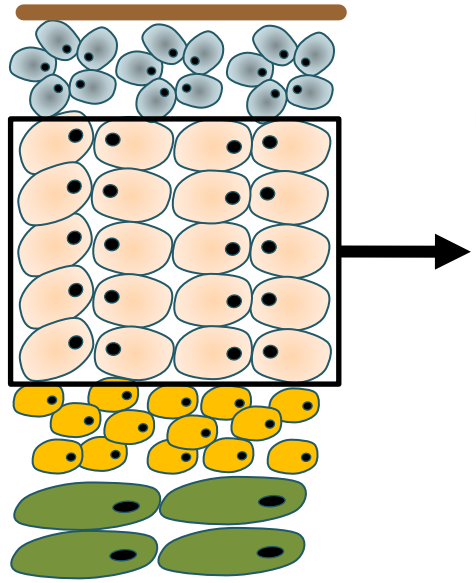
1. Aldosterone is produced in the zona glomerulosa. The enzyme 17α-hydroxylase is absent in this cortical layer, so progesterone is converted into 11-DOC by 21α-hydroxylase.
2. The enzyme aldosterone synthase catalyzes the conversion of 11-DOC via 2 separate intermediate mineralocorticoid products into aldosterone.
3. Aldosterone binds to the mineralocorticoid receptor inside the cells of the cortical collecting duct, where it stimulates the transcription of ENaC channels and Na^+/K^+ ATPases as well as the activity of ROMK channels. This increases Na^+ reabsorption and K^+ secretion.
4. Ang II stimulation of aldosterone secretion from the zona glomerulosa involves the IP_3 pathway and elevation of intracellular Ca. Calcium stimulates the secretion of aldosterone and increases the activity of side chain cleavage enzyme, StAR, and aldosterone synthase. Increases in extracellular K^+ stimulate aldosterone secretion by increasing intracellular Ca via depolarization of the membrane that opens voltage-gated Ca channels.

3. Describe the regulation of cortisol secretion including how exogenous steroid drugs affect adrenal cortical function.
4. Describe the target tissues and physiologic actions of cortisol.

Zona fasciculata

Glucocorticoids are steroids synthesized in the zona fasiculata

Glucocorticoids: Regulate GLUCOse and come from the CORTex



Cortisol (20 mg/24h) (87%)
Corticosterone (3 mg/24h) less potent

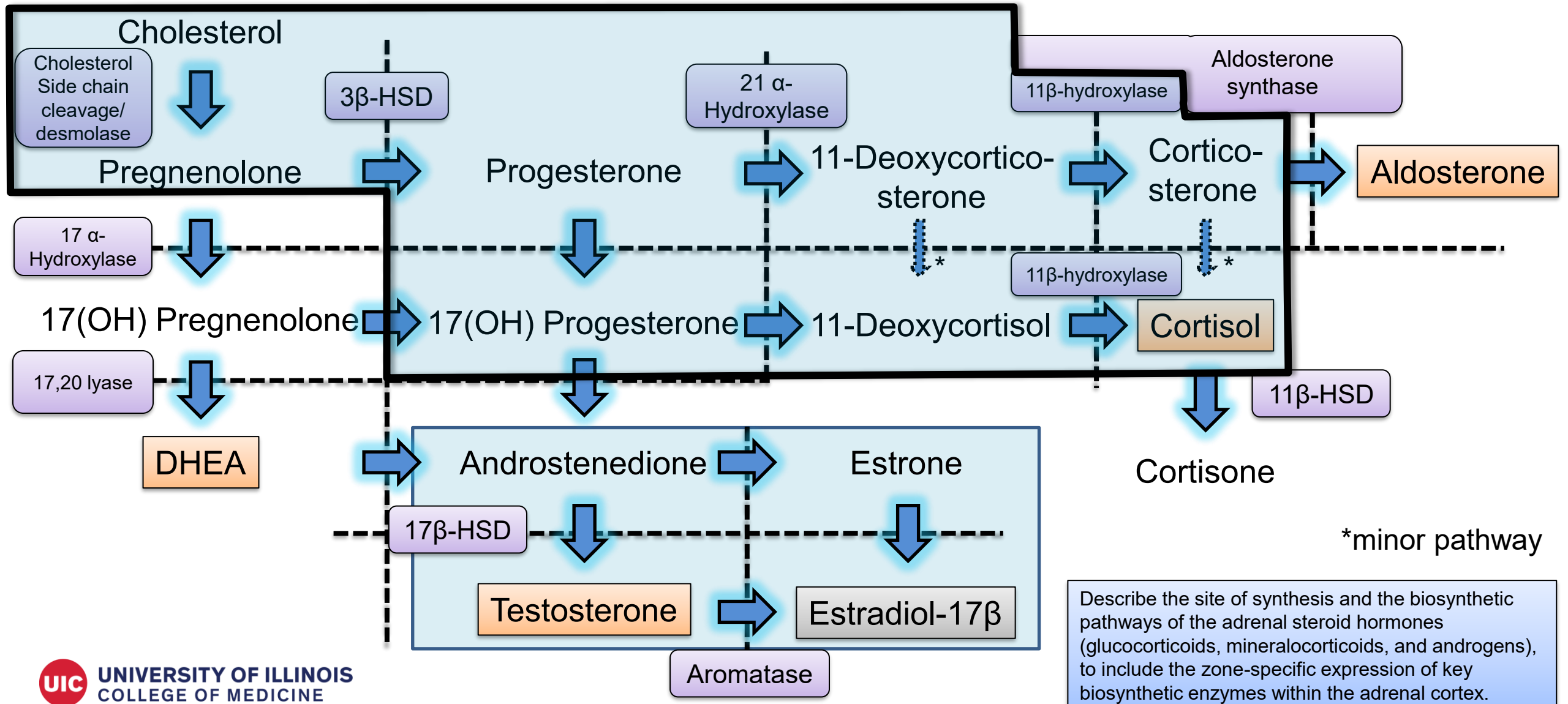
Under severe stress 300 to 400 mg of cortisol is secreted per day.

Systemic effects:

- Metabolic
- Psychiatric
- Immune
- Blood
- Skeletal

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

Steroid synthesis cheat sheet

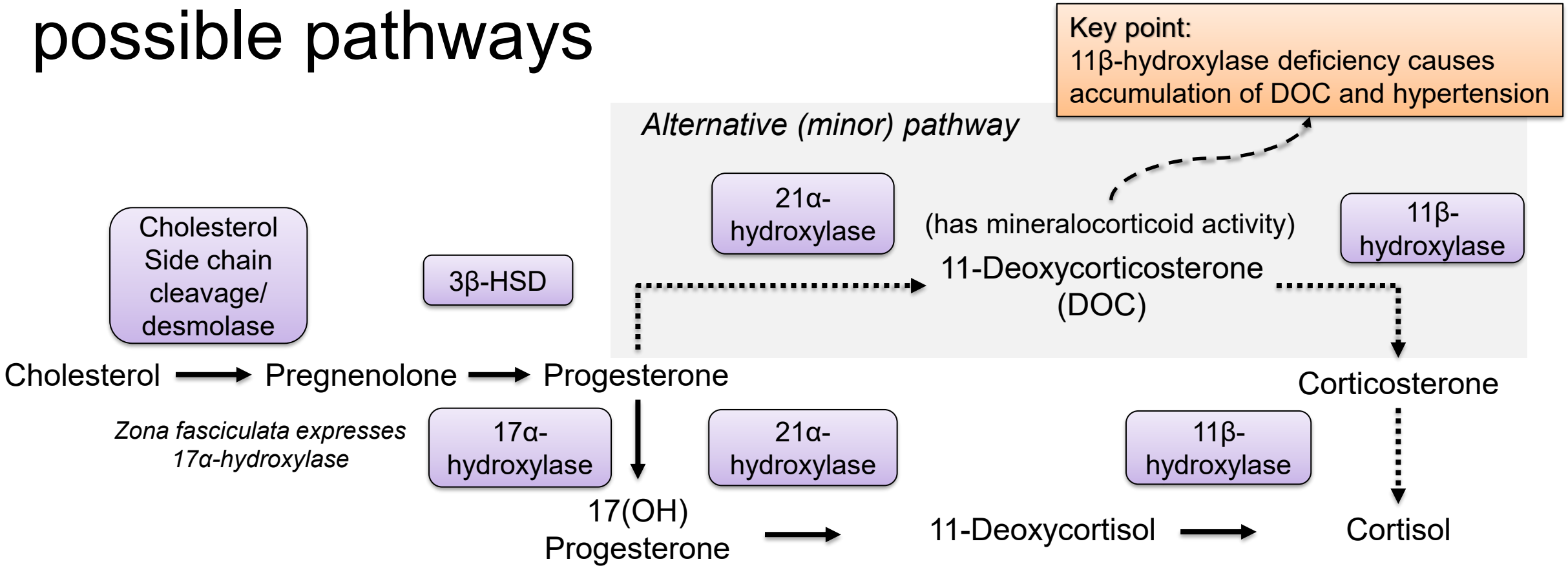


*minor pathway

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Biosynthesis of cortisol involves the enzymes 21α-hydroxylase and 11β-hydroxylase via two possible pathways



Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

Regulation of cortisol secretion

Circadian rhythm
Stress

Hypothalamus

CRH: Corticotropin Releasing Hormone

CRH

Anterior Pituitary

Corticotrope

POMC

ACTH

POMC: pro-opiomelanocortin

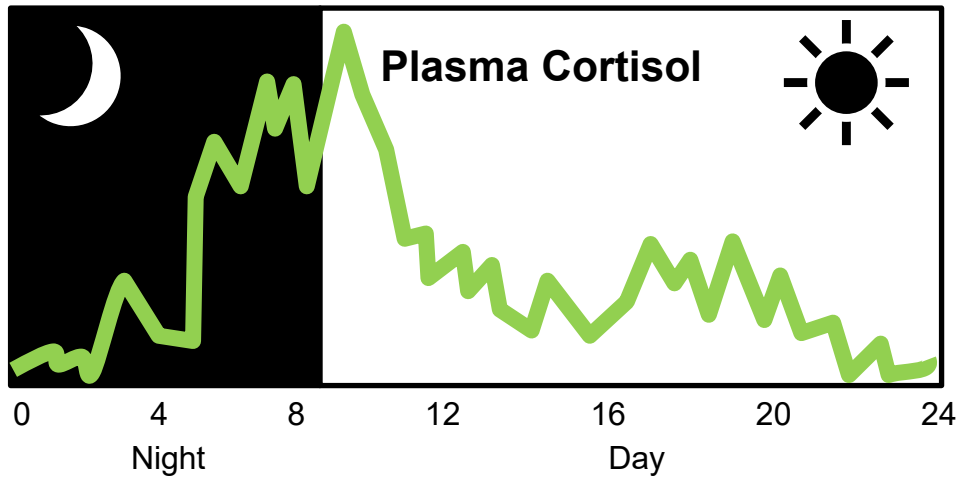
Adrenocorticotropic Hormone

ACTH has structural similarity to α-MSH (Melanocyte-Stimulating Hormone) produced in keratinocytes. Excess ACTH can cause skin darkening

ACTH

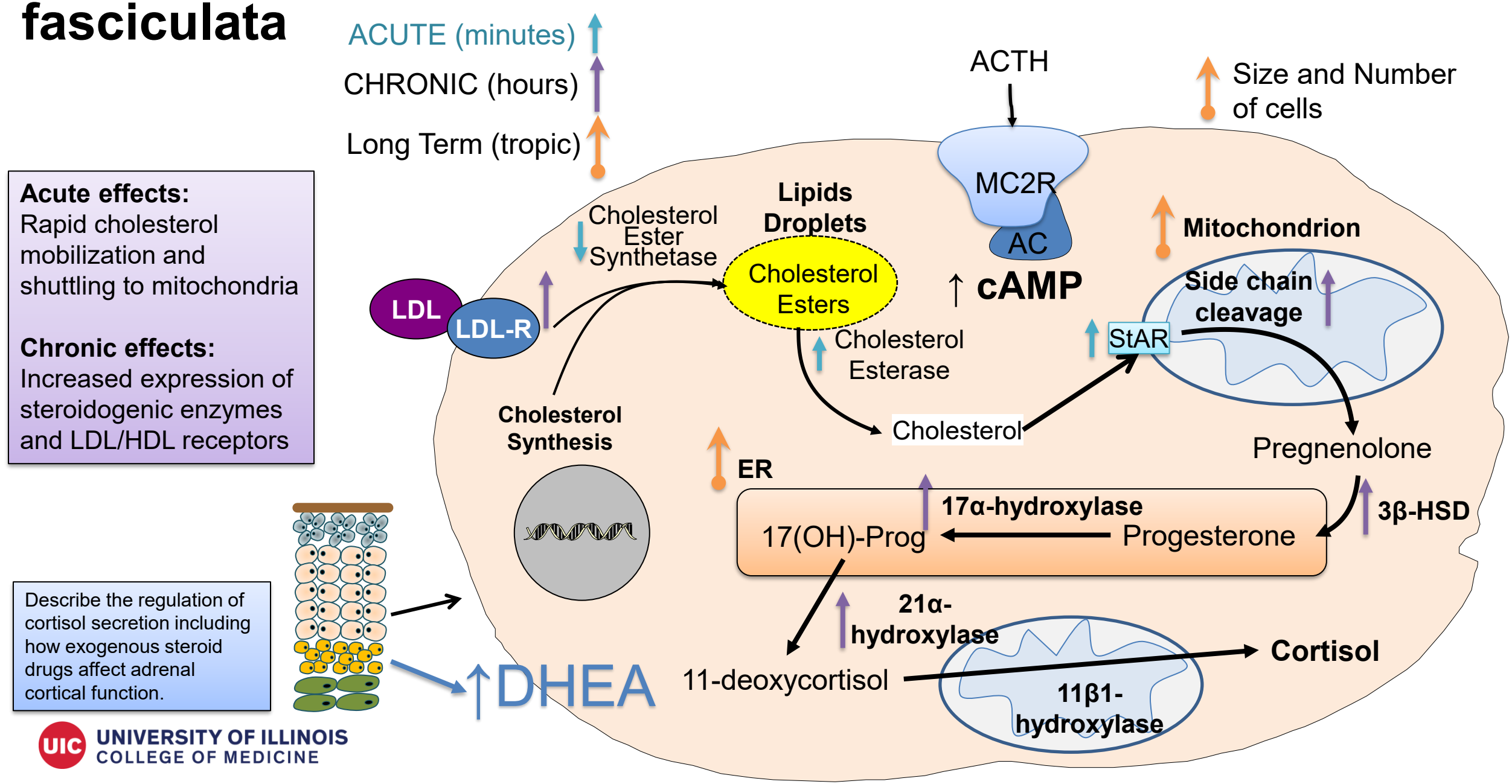
Zona fasciculata

Cortisol



Describe the regulation of cortisol secretion including how exogenous steroid drugs affect adrenal cortical function.

Acute, chronic and tropic effects of ACTH on the zona fasciculata



Acute effects:
Rapid cholesterol mobilization and shuttling to mitochondria

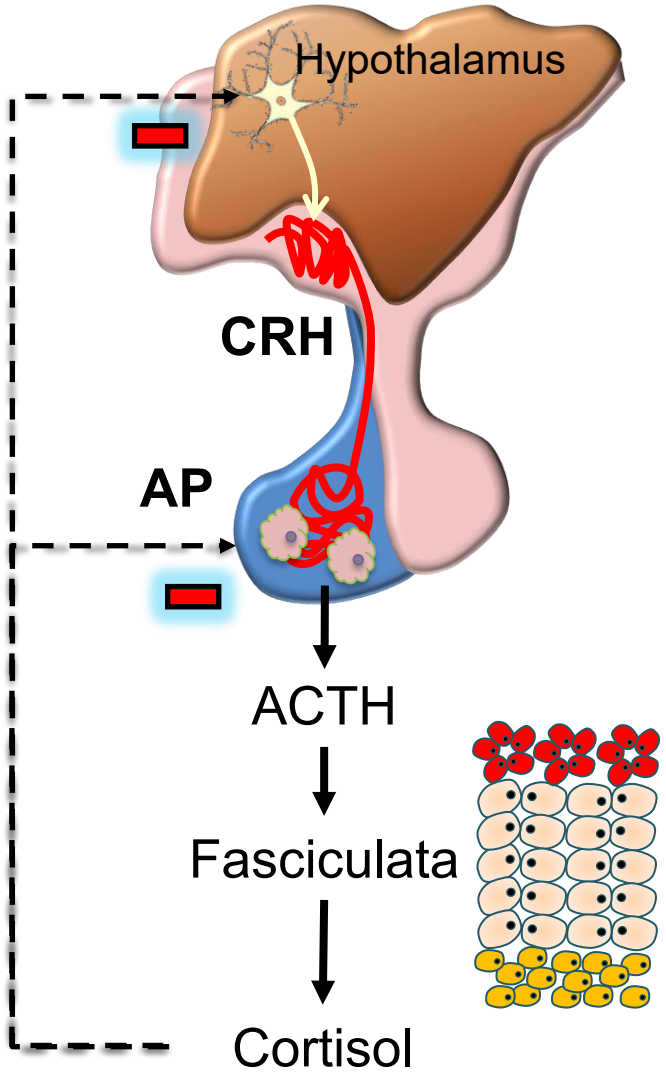
Chronic effects:
Increased expression of steroidogenic enzymes and LDL/HDL receptors

Describe the regulation of cortisol secretion including how exogenous steroid drugs affect adrenal cortical function.

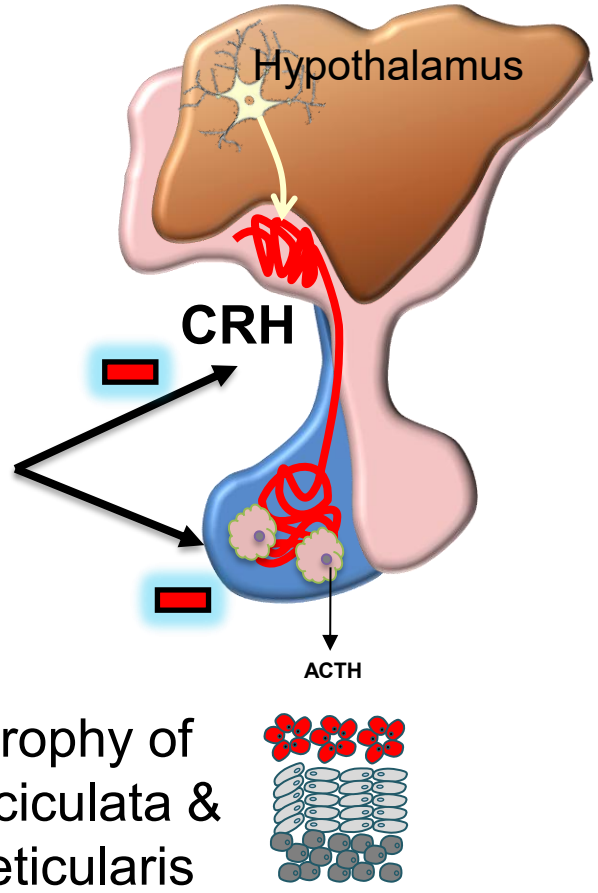
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Exogenous corticoids cause adrenal cortical atrophy

Key concept:
ACTH secretion has a trophic effect on the zona fasciculata & zona reticularis



Exogenous cortisol (prednisolone) > 3 weeks diminishes ACTH secretion

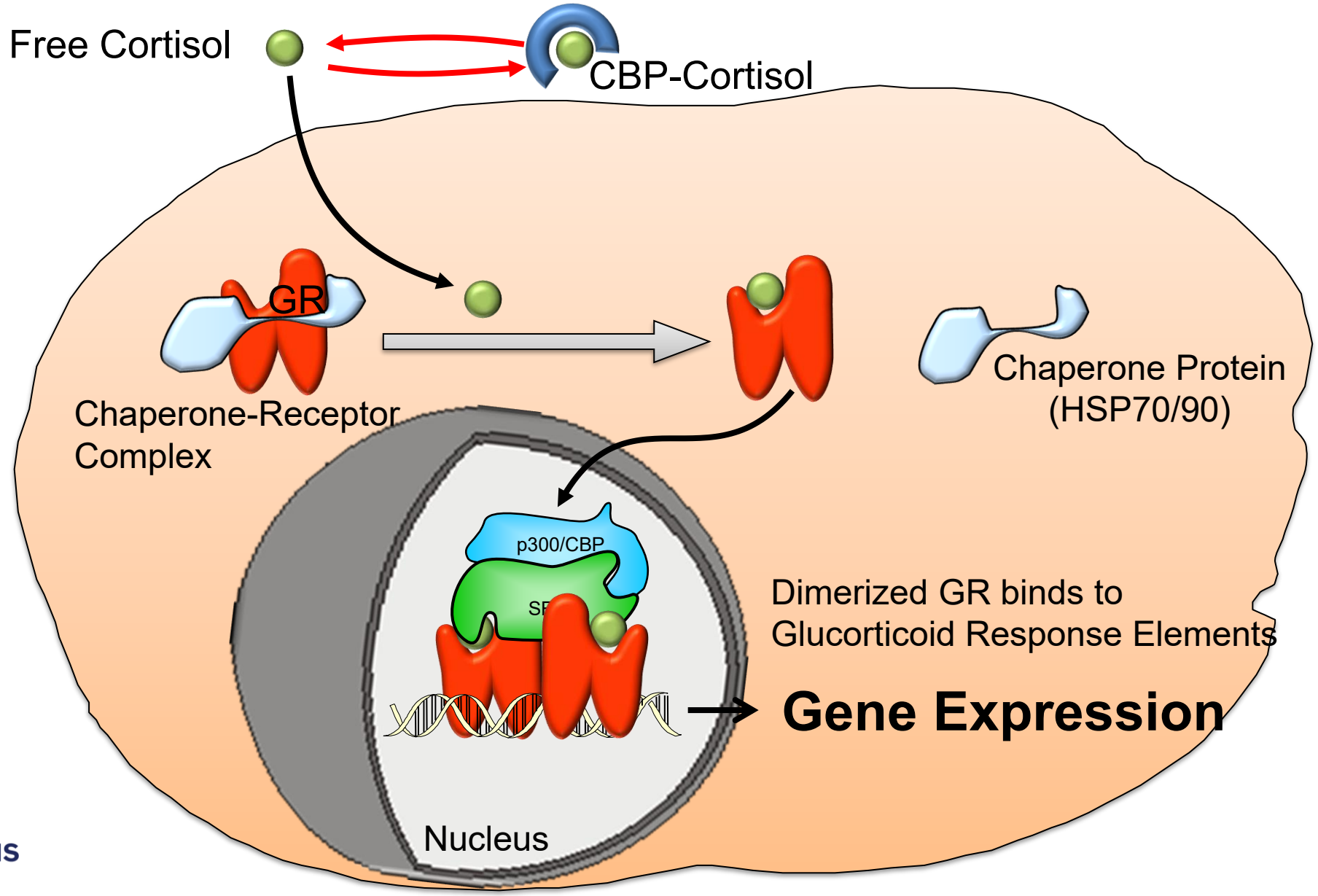


Withdraw gradually to allow rebuilding of the adrenal tissue

Describe the regulation of cortisol secretion including how exogenous steroid drugs affect adrenal cortical function.

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Cortisol acts through the glucorticoid receptor



Describe the cellular mechanism of action of adrenal cortical hormones.

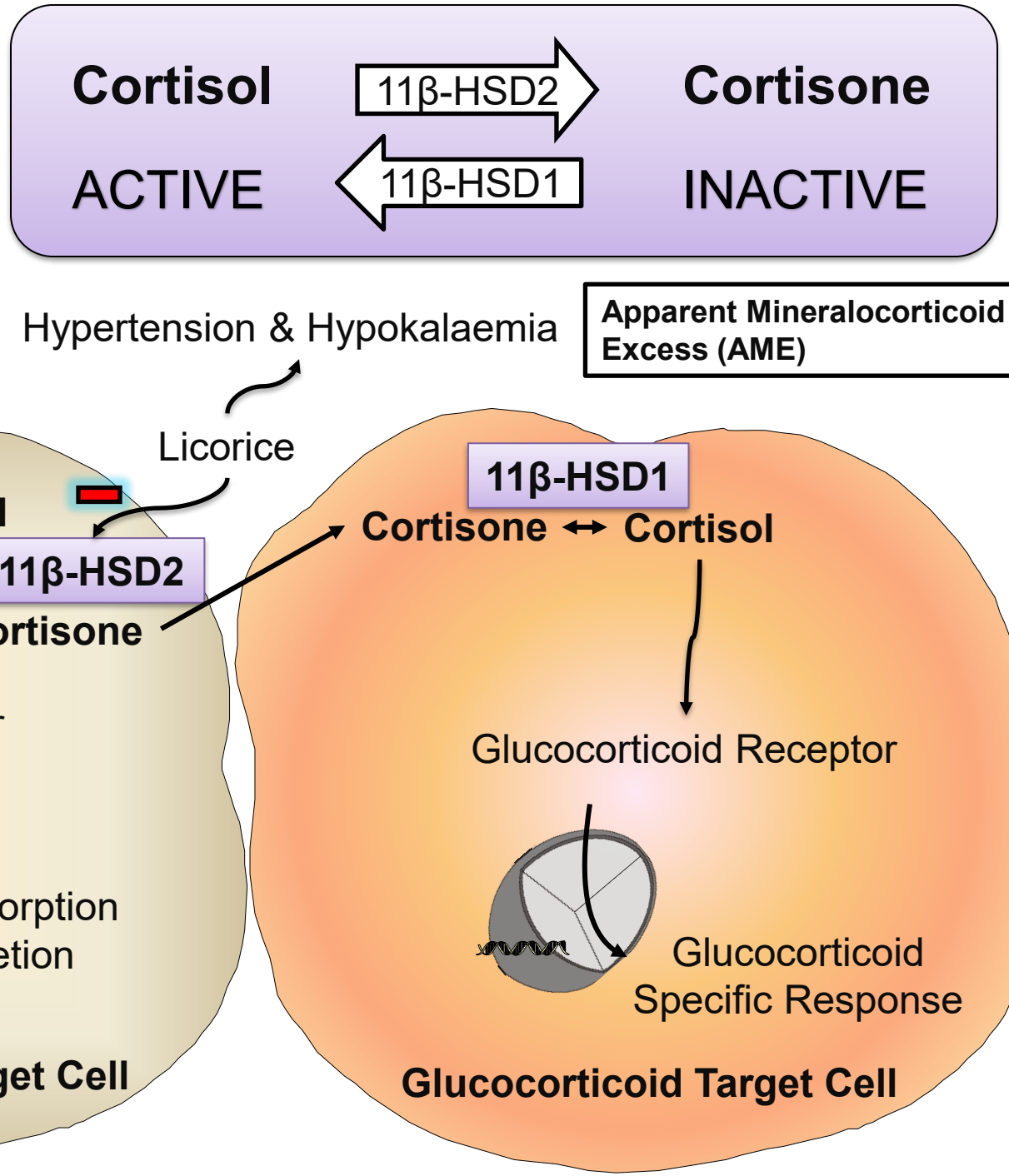
Relative Glucocorticoid and Mineralocorticoid Activity

Steroid	Plasma Concentration (μg/dL)	Glucorticoid Activity	Mineralocorticoid Activity
Cortisol	5-25 (8AM) 3-12 (4PM)	1	1
Corticosterone	1	0.5	1.5
Aldosterone	0.009	0.25	3000
11-Deoxycorticosterone (DOC)	0.07	0.2	100

Cortisol has notable mineralocorticoid activity and circulates in blood at about 1,000 times the concentration of aldosterone. So, why aren't we hypertensive?

Describe the target tissues and physiologic actions of cortisol.

Protection of the Mineralocorticoid Receptor by 11β-HSD2

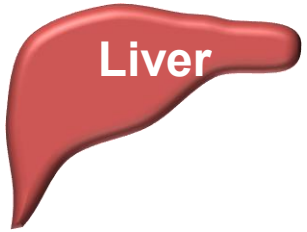


Key concept:
Cortisol activation of the mineralocorticoid receptor is prevented by conversion of cortisol into inactive cortisone

Describe the target tissues and physiologic actions of cortisol.

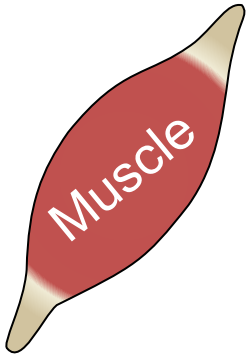
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Cortisol increases blood glucose



↑ Gluconeogenesis

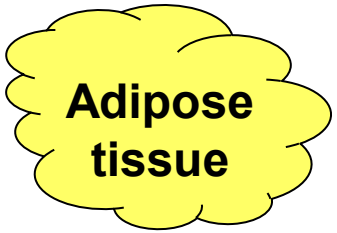
Increased PEPCK expression



Inhibits glucose uptake

Decreased protein synthesis

Increased proteolysis



Inhibits glucose uptake

Increased lipolysis

↑ **Glucose**

Note: Chronic excess cortisol synergizes with elevated insulin levels to cause lipogenesis and abdominal/visceral adiposity

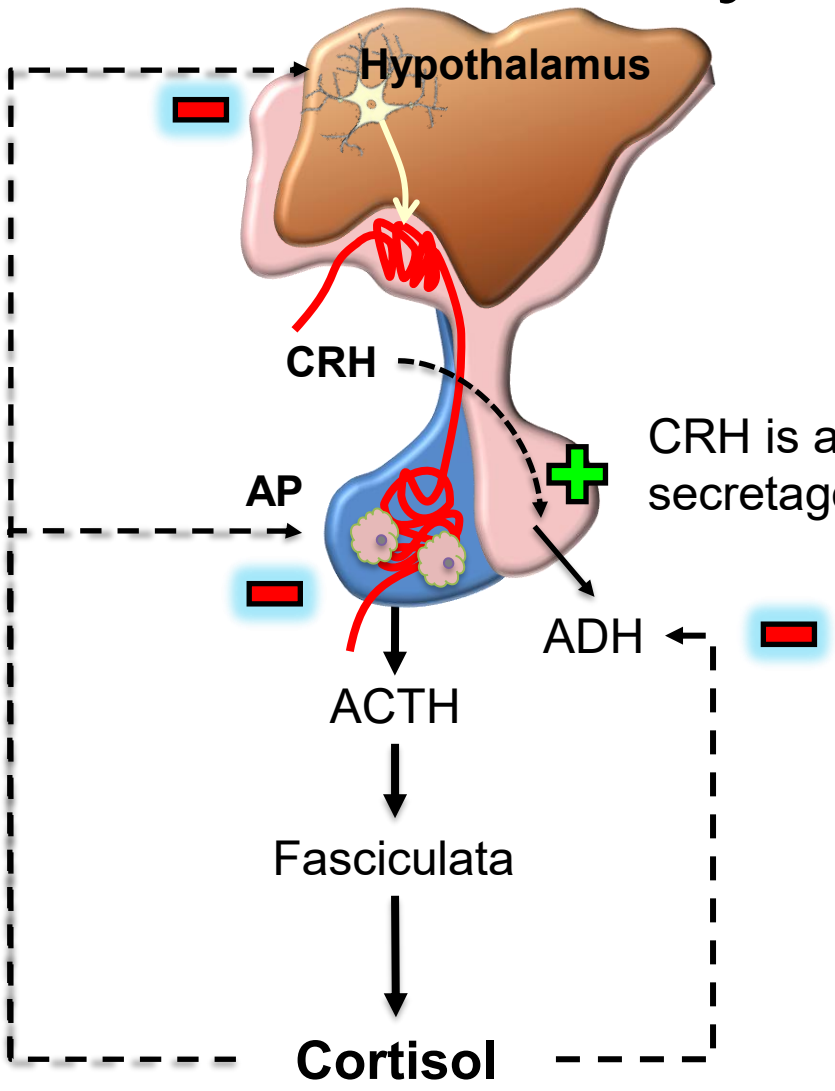
Describe the target tissues and physiologic actions of cortisol.

Systemic effects of cortisol

Organ system	Normal effect of cortisol	Effect of excess cortisol
Cardiovascular	Enhances action of catecholamines (increased adrenergic receptor expression) Erythropoietin synthesis	Polycythemia Hypertension (mineralocorticoid)
Reproductive system	Inhibits hormonal axis at hypothalamic and pituitary levels in response to stress	Infertility
Bone	Increases bone resorption (decreases serum Ca; inhibition of osteoblasts)	Osteoporosis
Connective tissue	Inhibition of fibroblast proliferation & collagen formation	Thinning of skin; easy bruising
Muscle		Weakness (proteolysis; hypokalemia)
GI tract	Trophic effect on gastric mucosa; increased gastric acid secretion	Increases appetite; ulcer development
Renal	Inhibits secretion and action of ADH Increased GFR	Mineralocorticoid activity
Psychiatric	Promotes normal well-being	Emotional liability & depression Insomnia

Describe the target tissues and physiologic actions of cortisol.

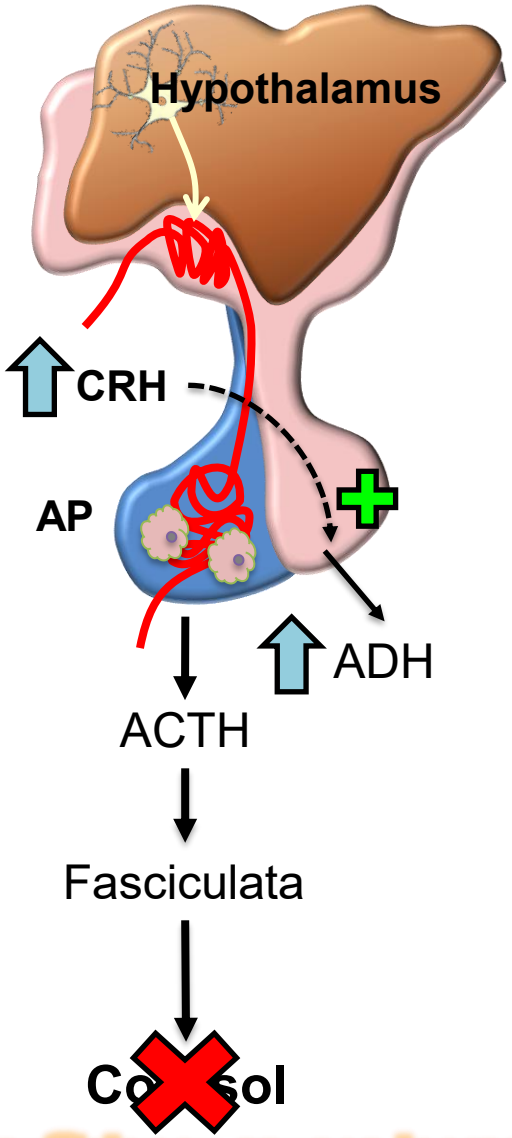
Adrenal insufficiency can cause hyponatremia



CRH is an ADH secretagogue

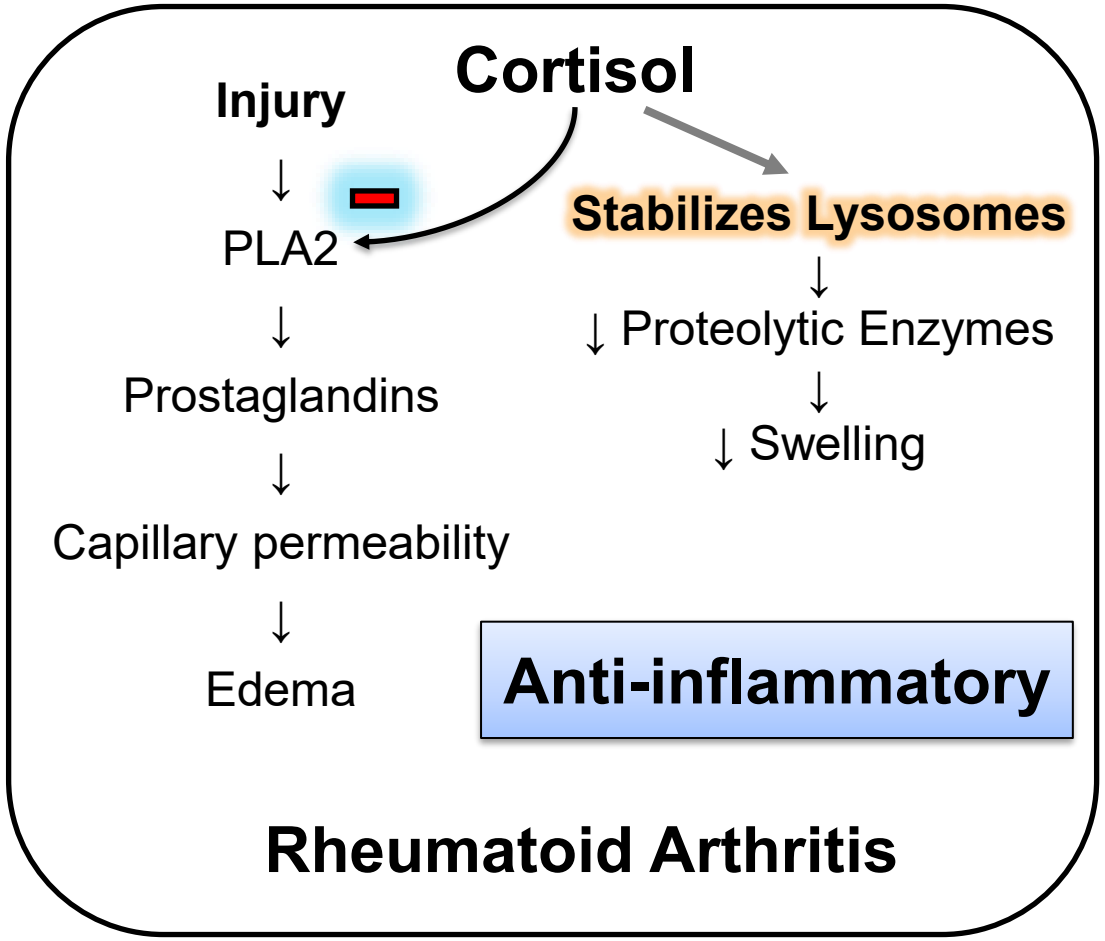
Describe the target tissues and physiologic actions of cortisol.

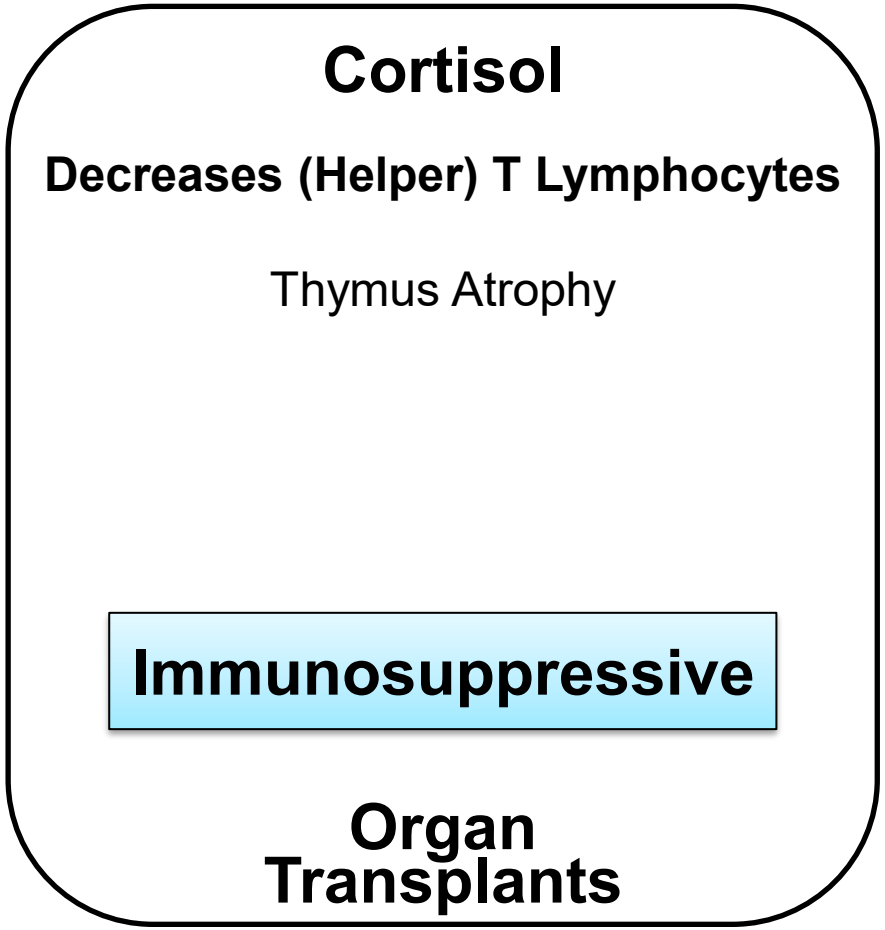
Cortisol inhibits ADH secretion



Adrenal insufficiency causes increased ADH levels due to loss of inhibition

Cortisol protects the body against uncontrolled inflammatory and immune responses





UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Describe the target tissues and physiologic actions of cortisol.

Q3: An individual with cortisol deficiency exhibits skin darkening. Which of the following is the most likely cause of the cortisol deficiency?

A. Primary adrenal insufficiency

B. Secondary adrenal insufficiency

C. Tertiary adrenal insufficiency

Q3 Answer: An individual with cortisol deficiency exhibits skin darkening. Which of the following is the most likely cause of the cortisol deficiency?

A. Primary adrenal insufficiency

B. Secondary adrenal insufficiency

C. Tertiary adrenal insufficiency

Skin darkening with cortisol deficiency is due to excess ACTH secretion. In primary adrenal insufficiency, the adrenal cortex is unable to produce cortisol, which leads to high ACTH levels.
In secondary adrenal insufficiency, the low cortisol levels are due to diminished pituitary function, leading to low ACTH levels.
In tertiary adrenal insufficiency, low cortisol production is due to low hypothalamic secretion of CRH, which leads to low ACTH secretion from the pituitary.
In secondary and tertiary adrenal sufficiency, the ACTH levels are low, which therefore would not result in skin darkening.

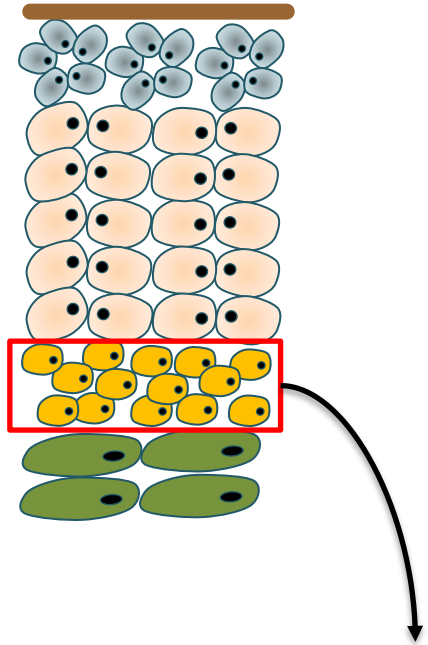
Summary of zona fasciculata

1. The zona fasciculata produces the glucocorticoids cortisol and corticosterone.
2. The biosynthetic pathway uses the enzymes 3β-HSD, 17α-hydroxylase, 21α-hydroxylase and 11β-hydroxylase. An alternative minor pathway that bypasses 17α-hydroxylase generates 11-DOC, which has mineralocorticoid activity.
3. Cortisol secretion exhibits diurnal variation due to a circadian rhythm, with highest amounts of cortisol secretion in the morning. Stress (neurogenic and physiological) also stimulates cortisol production.
4. The hypothalamus releases CRH, which acts on the corticotropes of the anterior pituitary to release ACTH (synthesized from POMC). ACTH stimulates the zona fasciculata to secrete cortisol. Cortisol reduces the secretion of CRH and ACTH by negative feedback.
5. ACTH acutely increases the shuttling of cholesterol precursors through the biosynthetic pathway in the zona fasciculata. Chronically it increases the expression of biosynthetic enzymes and also has a trophic effect on the zona fasciculata and zona reticularis.
6. Cortisol binds to the glucocorticoid receptor, which stimulates gene transcription inside cells.
7. Cortisol has mineralocorticoid activity, but conversion to the inactive cortisone by 11β-HSD2 prevents cortisol binding to the mineralocorticoid receptor. The enzyme 11β-HSD1 converts cortisone back to cortisol.
8. Cortisol increases blood glucose by stimulating gluconeogenesis and by reducing glucose uptake into muscle and adipose tissue. It has many systemic effects, including anti-inflammatory and immunosuppressive effects.

7. Describe the regulation of adrenal androgen production.

Zona reticularis

The zona reticularis synthesizes adrenal androgens



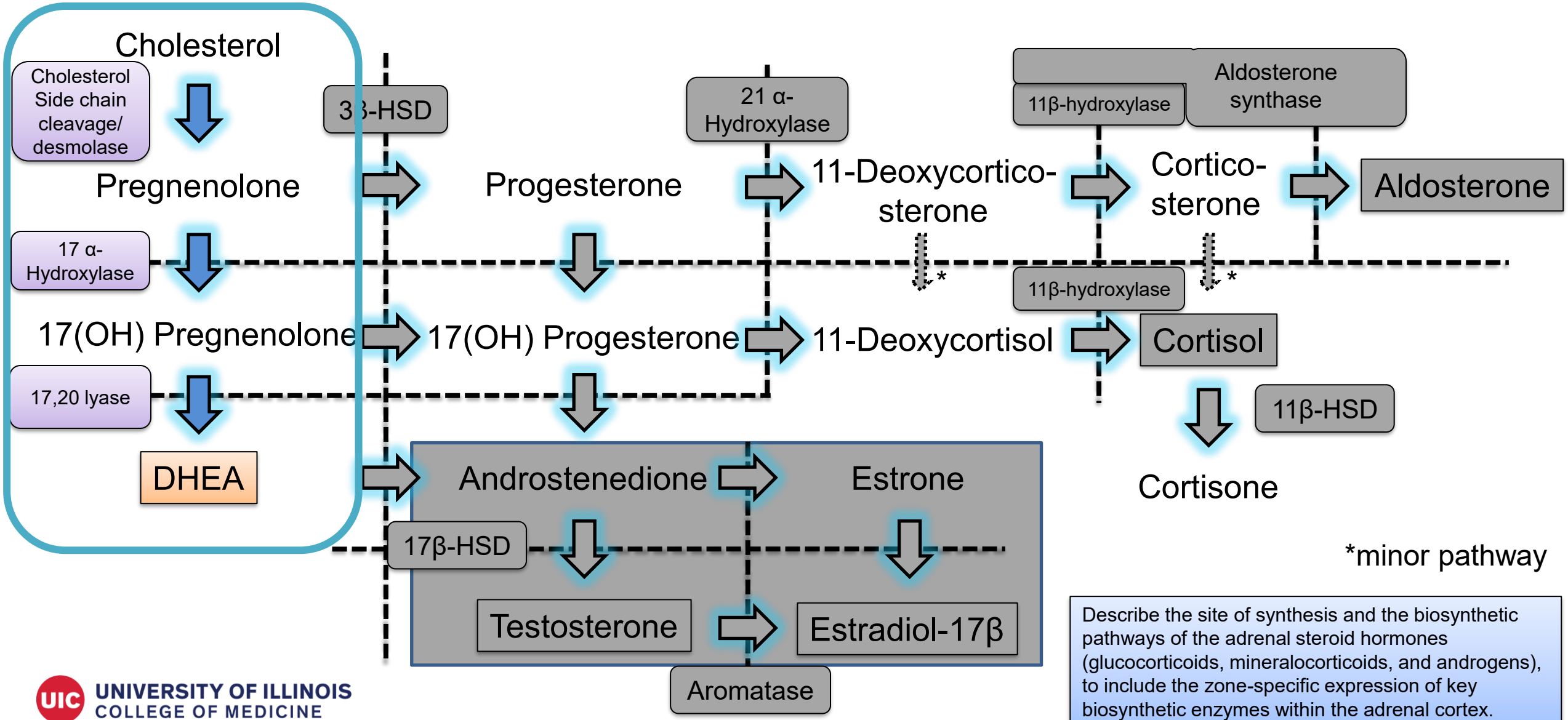
- Contributes to the development of axillary and pubic hair in children >8 years old
- Negligible contribution to active androgens in males
- ~50% of circulating androgens in females
  - Important for axillary & pubic hair growth and Libido
  - Excess can cause masculinization & hirsuitism

Dehydroepiandrosterone (DHEA)
DHEA Sulfate
Androstenedione

Potent androgens (testosterone) are produced by peripheral conversion of DHEAS and androstenedione

Describe the physiologic role of adrenal androgens

Steroid synthesis cheat sheet

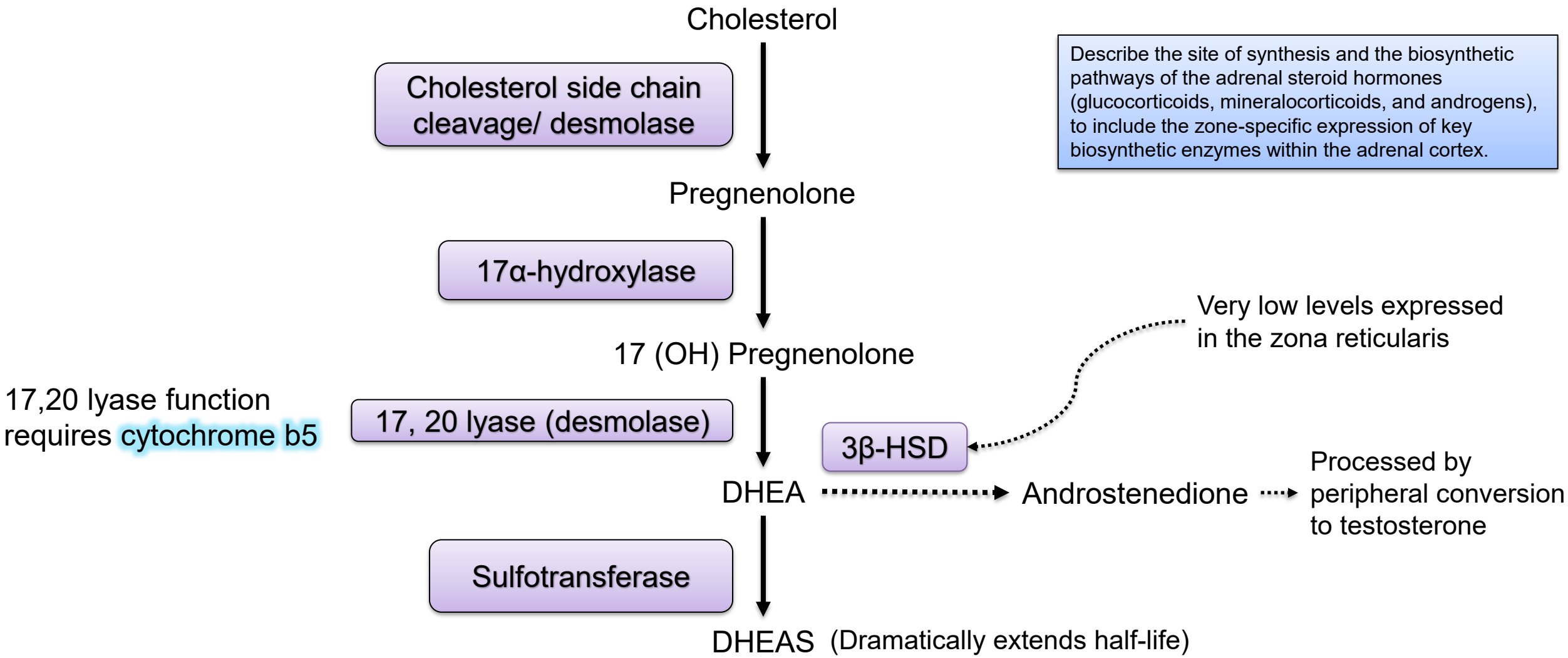


*minor pathway

Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

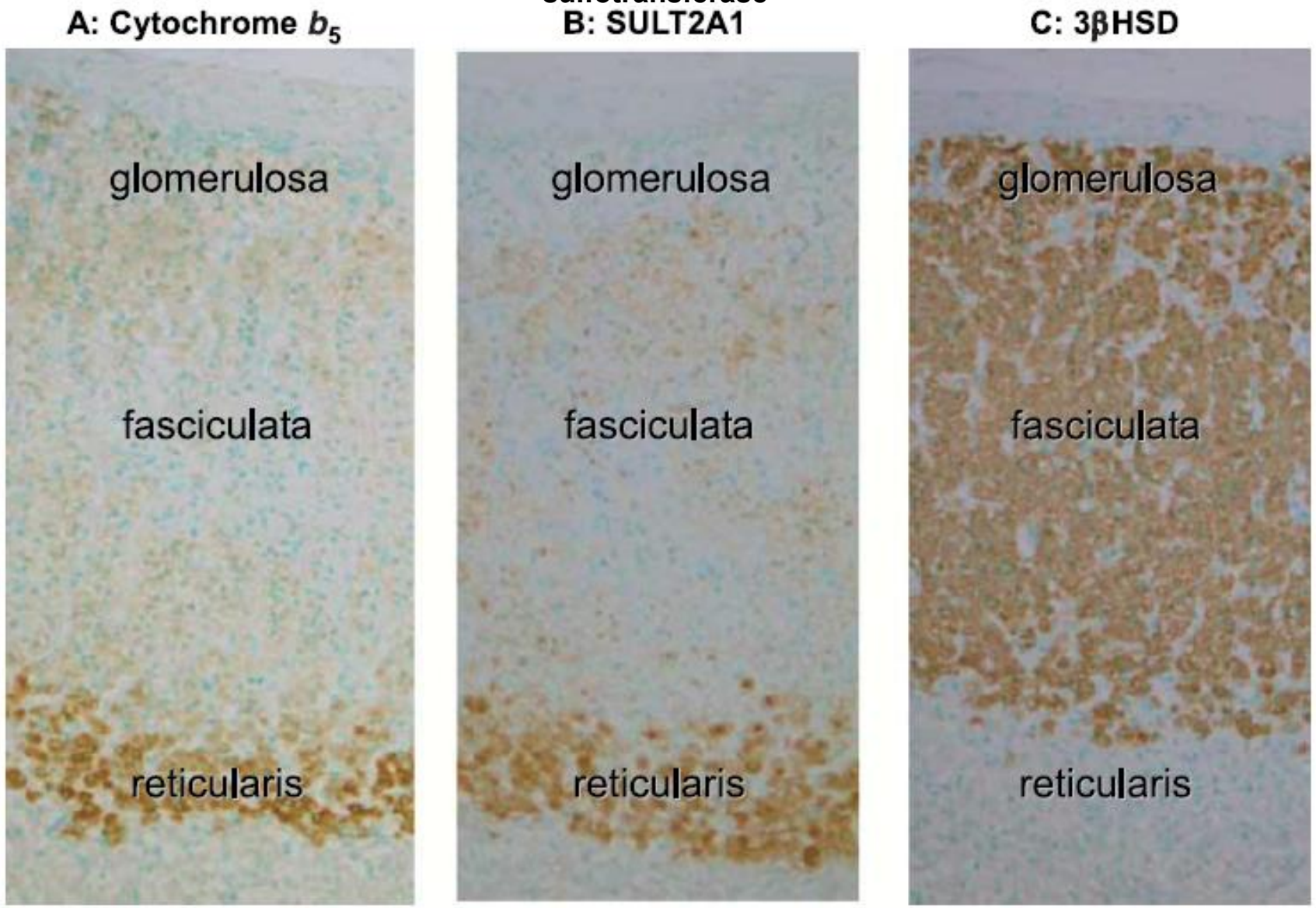
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Biosynthetic pathways for DHEA and DHEAS



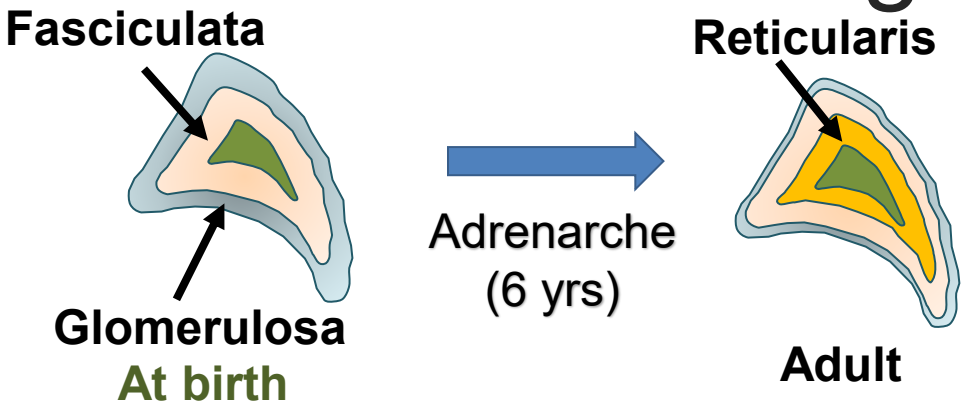
Describe the site of synthesis and the biosynthetic pathways of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens), to include the zone-specific expression of key biosynthetic enzymes within the adrenal cortex.

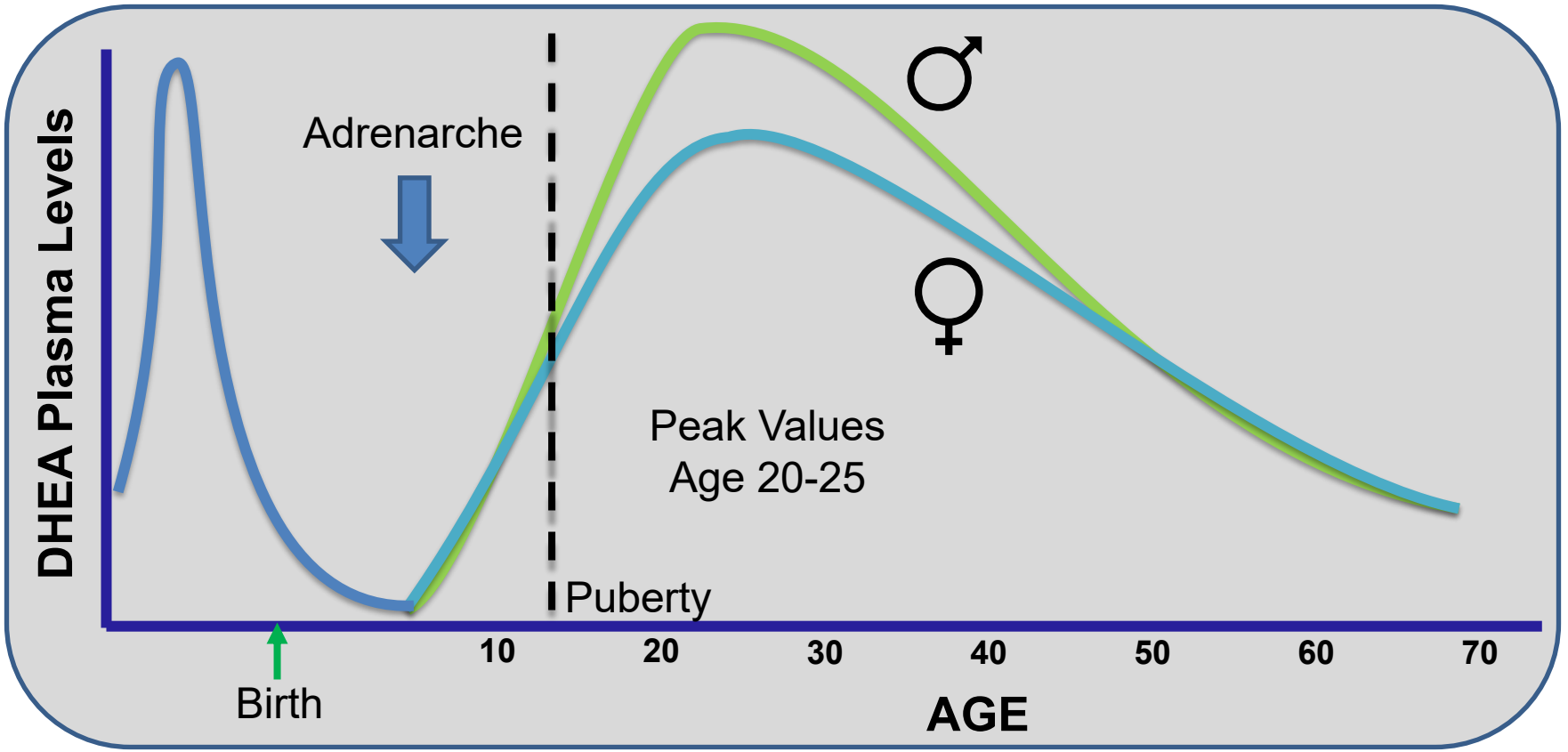
Immunohistochemistry of Adrenal Cortex In Early Adrenarche



Clinical Endocrinology (2004) **60**, 288–296

Age and Adrenal Androgens

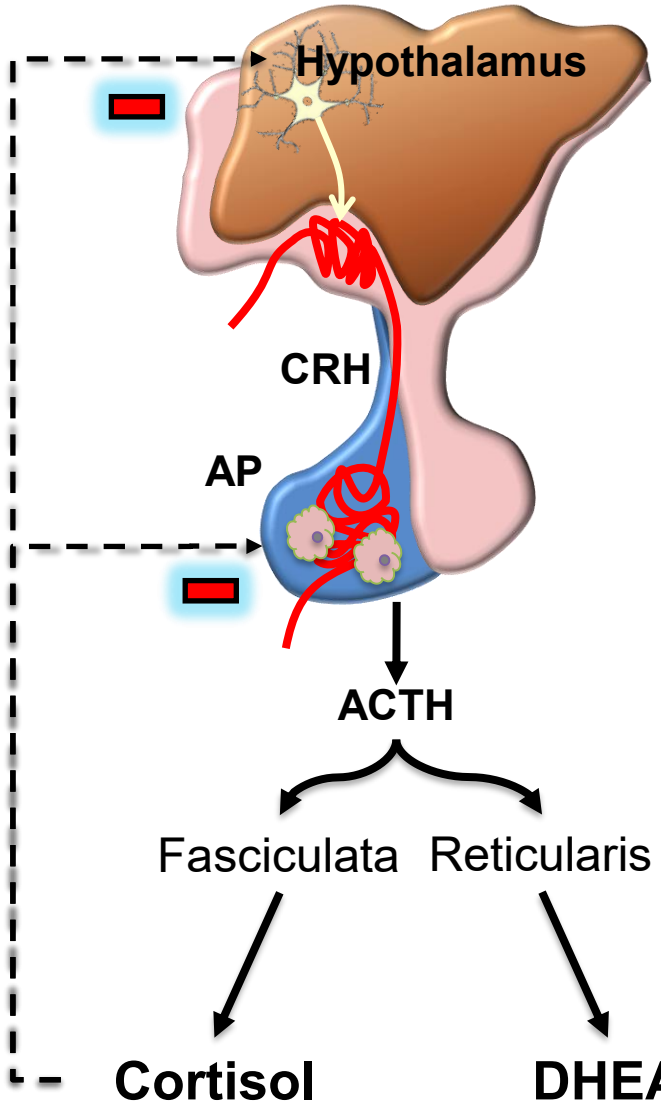




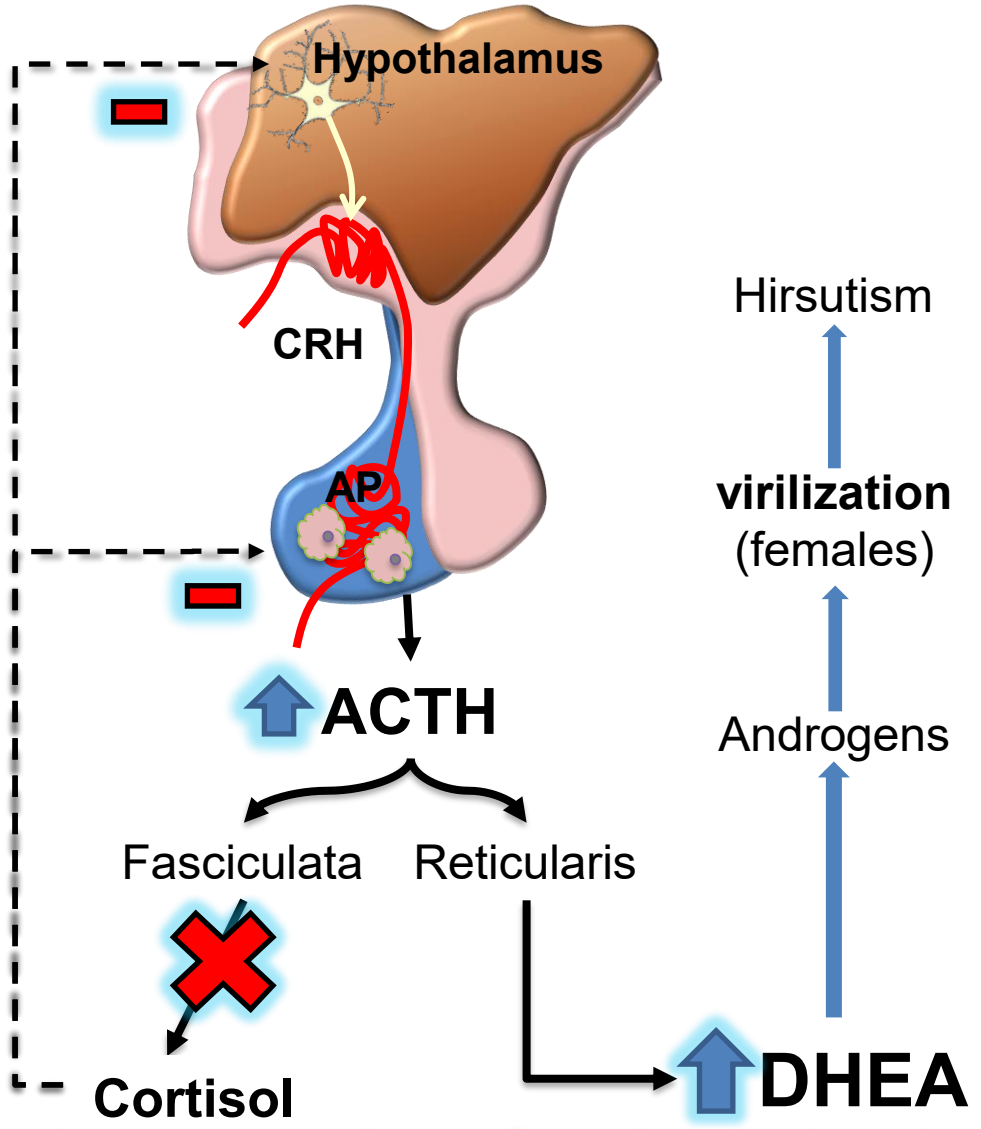
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Primary regulation of zona reticularis function by ACTH

Describe the effect of ACTH on adrenal androgen production.



Normal



Impairment of cortisol synthesis (e.g. 21α-hydroxylase deficiency) leads to excess androgen production

Q4: Which of the following enzymes is involved in the production of DHEA but not cortisol or aldosterone?

A. Cholesterol side chain cleavage
B. 17α-hydroxylase
C. 17,20 lyase
D. 3β-hydroxysteroid dehydrogenase
E. 21α-hydroxylase

Q4 Answer: Which of the following enzymes is involved in the production of DHEA but not cortisol or aldosterone?

A. Cholesterol side chain cleavage

B. 17α-hydroxylase

C. 17,20 lyase

D. 3β-hydroxysteroid dehydrogenase

E. 21α-hydroxylase

Cholesterol side chain cleavage enzyme and 17α-hydroxylase are common to the synthesis of DHEA, aldosterone, and cortisol. 3β-HSD and 21$^{\alpha}$-hydroxylase are required for the synthesis of cortisol and aldosterone, but not DHEA. 17,20 lyase is the only enzymatic process involved in DHEA synthesis but not synthesis of cortisol or aldosterone.

Summary of zona reticularis

1. The zona reticularis produces dehydroepiandrosterone, which is sulfated to prolong half life, and small quantities of androstenedione. These are converted in peripheral tissues to testosterone and estradiol.
2. The synthesis of DHEA uses the Δ5 pathway, since there are very low levels of 3β-HSD in the zona reticularis. The presence of cytochrome b5 stimulates the 17,20 lyase reaction.
3. Adrenarche occurs around the age of 6 after the development of the zona reticularis. Peak adrenal androgen production is around 20-25 years of age.

Compare and contrast Congenital Adrenal Hyperplasia (CAH) caused by deficiency of: 21α-hydroxylase, 11β-hydroxylase & 17α-hydroxylase.

Congenital Adrenal Hyperplasia (CAH)

Causes of Congenital Adrenal Hyperplasia

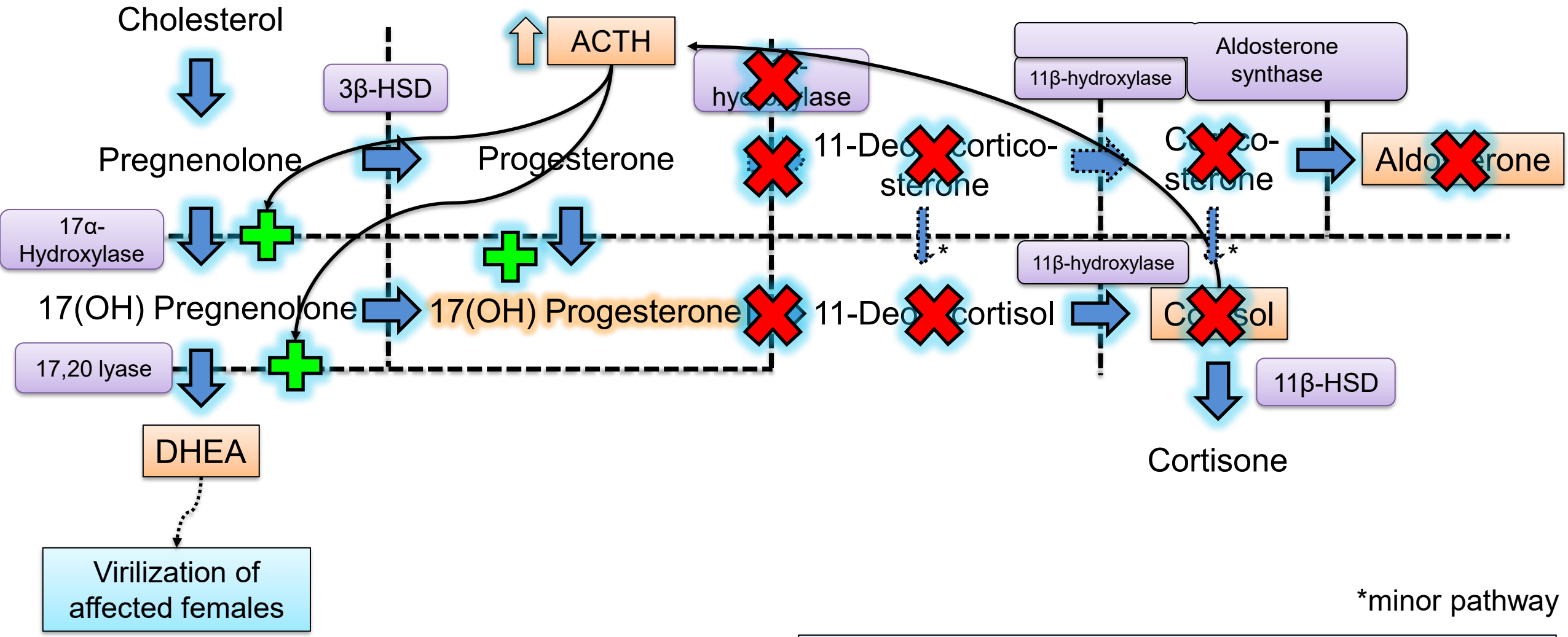
- Genetic disorders causing defects in biosynthetic enzymes that result in decreased cortisol synthesis in the adrenal cortex.
  - 21α-hydroxylase deficiency (major form)
  - 11β-hydroxylase deficiency (rare)
  - 17α-hydroxylase deficiency (rare)
  - Other rare forms (not mentioned here)

- Aldosterone production can also be affected depending on the enzymatic deficiency

- Adrenal androgen production is increased (21α-hydroxylase & 11β-hydroxylase deficiency) or absent (17α-hydroxylase deficiency) which can lead to ambiguous genitalia or disorders of sexual development

- Severity of disease is variable and depends on the degree of enzyme deficiency

Classic 21α-hydroxylase deficiency leads to congenital adrenal hyperplasia with salt-*wasting* and excess adrenal androgens



UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

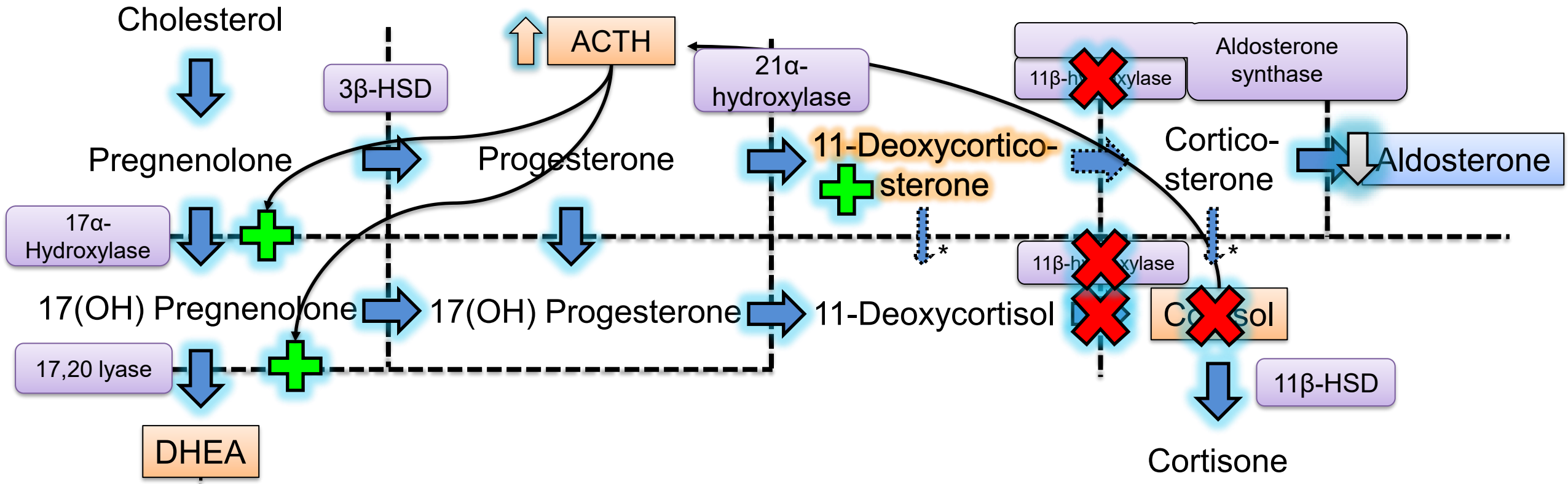
Compare and contrast Congenital Adrenal Hyperplasia (CAH) caused by deficiency of: 21α-hydroxylase, 11β-hydroxylase & 17α-hydroxylase.

Clinical features of CAH caused by 21α-hydroxylase deficiency

- 95% of CAH results from 21α-hydroxylase deficiency
- Very high serum levels of 17-hydroxyprogesterone are diagnostic
- Risk for adrenal crisis (acute adrenal insufficiency)
- Lack of mineralocorticoid leads to salt-wasting that can lead to hypotension and hypovolemia, accompanied by hyperkalemia and metabolic acidosis
- Atypical genitalia
- Risk for advanced bone age, premature epiphyseal closure and short stature due to high levels of androgens

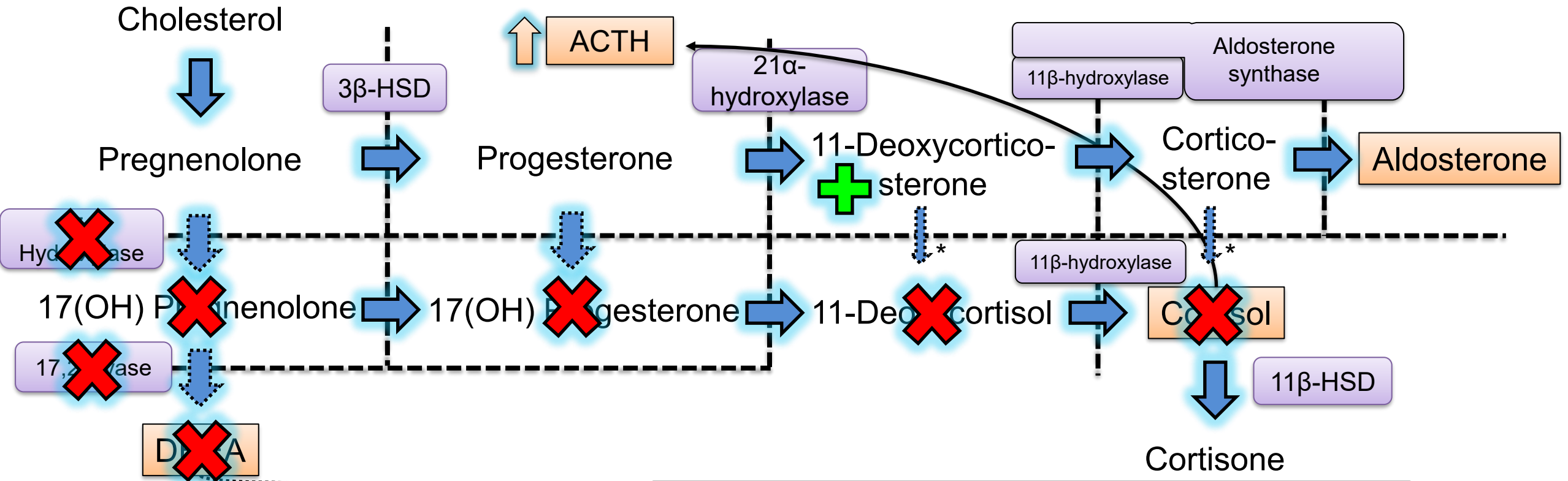
*Note: there is also nonclassic CAH caused by late onset 21α-hydroxylase deficiency that presents with increased adrenal androgen production without atypical genitalia

11β-hydroxylase deficiency leads to congenital adrenal hyperplasia with *salt-retention* and excess adrenal androgens



UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

17α-hydroxylase deficiency leads to congenital adrenal hyperplasia with hypertension and lack of adrenal androgen production



Primary amenorrhea & absence of secondary sexual characteristics
Blind vaginal pouch (46,XY)
Small uterus and ovaries (46,XX)

Elevated DOC/Corticosterone:Cortisol ratio as well as low androgens and estrogens is diagnostic

*minor pathway

Compare and contrast Congenital Adrenal Hyperplasia (CAH) caused by deficiency of: 21α-hydroxylase, 11β-hydroxylase & 17α-hydroxylase.

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Comparing the features of different forms of CAH

Enzyme deficiency	Mineralocorticoid production	Androgen production	Clinical features	Diagnostics
21α-hydroxylase	Decreased	Increased	• Atypical genitalia • Adrenal crisis (nausea, vomiting, hypotension, hypovolemic shock) • Hyperkalemia (± hyponatremia) • Metabolic acidosis • Hypoglycemia	• Low cortisol • Very high serum 17(OH) Progesterone • High androgens • Low aldosterone
11β-hydroxylase	Increased	Increased	• Atypical genitalia • Hypertension • Hypokalemia • Premature adrenarche and advanced bone age if not treated at birth	• Low cortisol • High serum 11-deoxycortisol & DOC • High androgens • Low aldosterone
17α-hydroxylase	Increased	Decreased	• Hypertension • Primary amenorrhea • Absence of body hair & secondary sexual characteristics • Blind vaginal pouch (46,XY) • Small uterus & ovaries (46,XX)	• Low cortisol • Elevated serum DOC & Corticosterone • Low androgens and estrogens • Low aldosterone

Source: UpToDate

Compare and contrast Congenital Adrenal Hyperplasia (CAH) caused by deficiency of: 21α-hydroxylase, 11β-hydroxylase & 17α-hydroxylase.

An infant (46,XX) is born with an enlarged clitoris that has the appearance of a micropenis. The infant has a very high serum level of 17-hydroxyprogesterone. Which of the following would also likely be present in this patient?

A. High serum cortisol levels

B. High serum DOC levels

C. Low extracellular fluid volume

D. Low serum DHEAS levels

An infant (46,XX) is born with an enlarged clitoris that has the appearance of a micropenis. The infant has a very high serum level of 17-hydroxyprogesterone. Which of the following would also likely be present in this patient?

A. High serum cortisol levels

B. High serum DOC levels

C. Low extracellular fluid volume

D. Low serum DHEAS levels

This infant exhibits features suggestive of CAH due to 21α-hydroxylase deficiency, which results in elevated levels of 17-hydroxyprogesterone and increased adrenal androgen production. Due to the lack of 21α-hydroxylase, the infant is unable to synthesize mineralocorticoids, such as aldosterone or DOC. Serum cortisol levels will also be low, as 21α-hydroxylase is part of the synthetic pathway for cortisol. The lack of aldosterone will lead to low extracellular fluid volume and potentially hypotension.

Summary: CAH

1. Impairment of cortisol production by 21α-hydroxylase or 11β-hydroxylase deficiency leads to loss of feedback on the corticotropes in the anterior pituitary and consequently excess ACTH production. This expands the zona reticularis and leads to the over-production of adrenal androgens and atypical genitalia in newborns.
2. 11β-hydroxylase deficiency leads to salt retention due to the production of 11-DOC, which has mineralocorticoid activity. In contrast, 21α-hydroxylase deficiency leads to salt-wasting, as steroid precursors with mineralocorticoid activity are not manufactured.
3. 17α-hydroxylase deficiency leads to lack of cortisol and adrenal androgen production as well as hypertension from increased mineralocorticoid production.

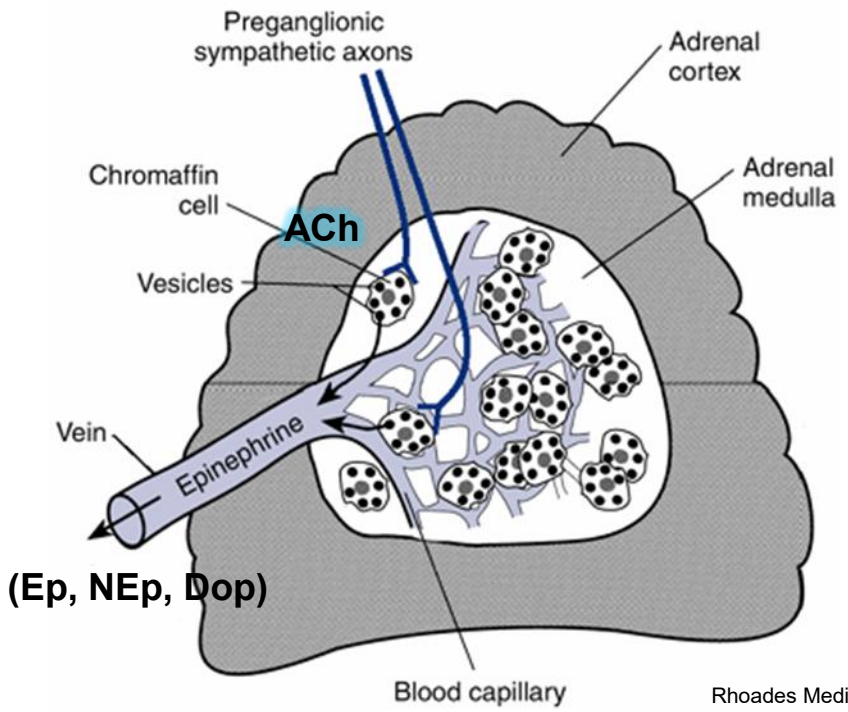
8. Describe the chemical nature of catecholamines, their biosynthesis, and stimulus for secretion.
9. Describe the influence of cortisol on the adrenal medulla and explain its physiological significance during stress.

Adrenal medulla

The adrenal medulla is a modified sympathetic ganglion

In the **adrenal gland**, the sympathetic preganglionic neurons travel directly to the adrenal medulla, where they release **ACh** (like a modified sympathetic ganglion).

ACh stimulates *Chromaffin cells* to release **Epinephrine (80%), Norepinephrine (20%), & Dopamine** (trace amounts) directly into the bloodstream. These products dissolve in plasma, so no carrier protein is needed.



Rhoades Medical Physiology, 3rd ed

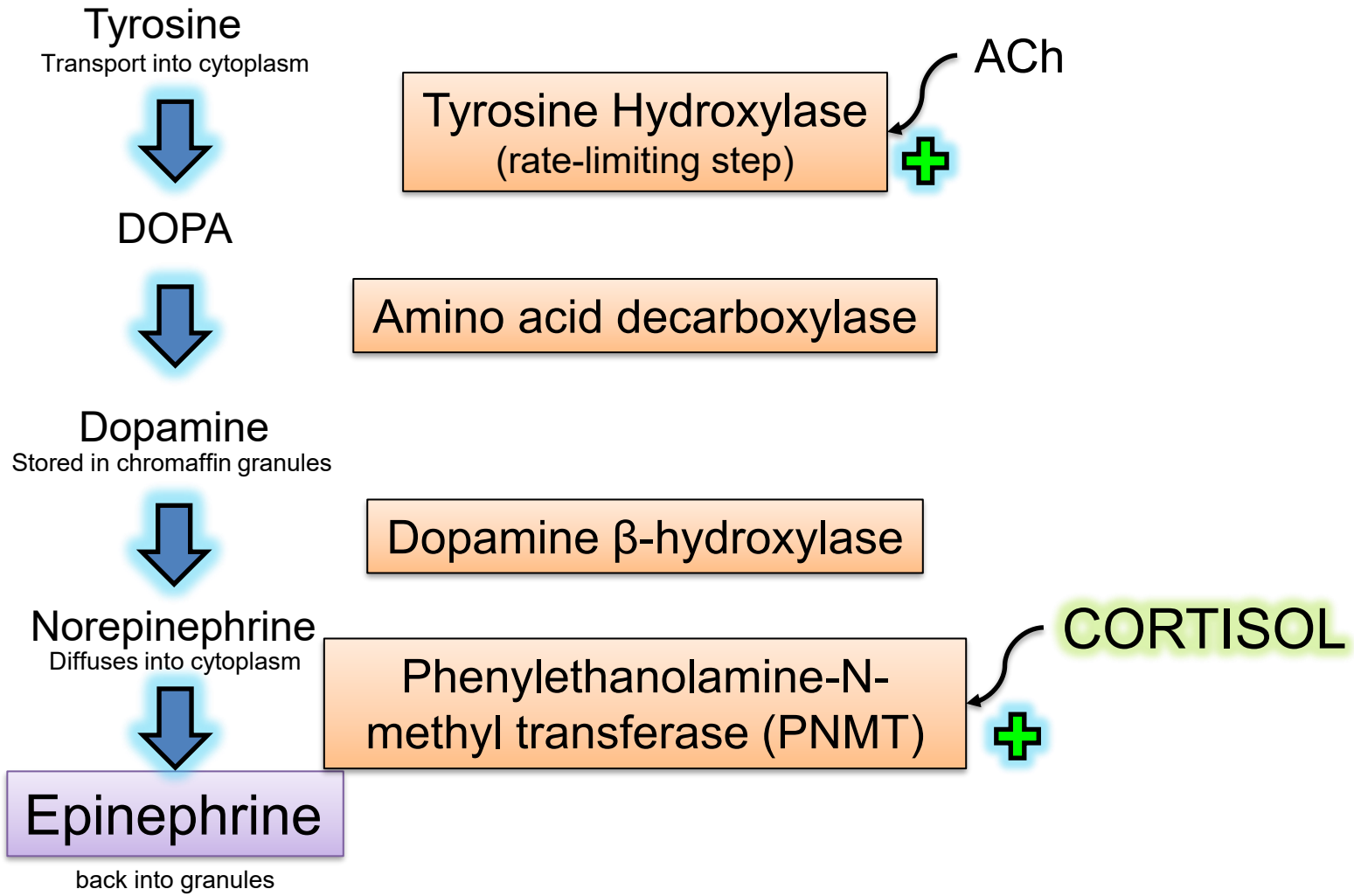
Describe the chemical nature of catecholamines, their biosynthesis, and mechanism of transport within the blood.

Synthesis of epinephrine in the adrenal medulla is regulated by cortisol

Key point:
Virtually all circulating epinephrine comes from the adrenal medulla due to chromaffin cell-specific expression of PNMT

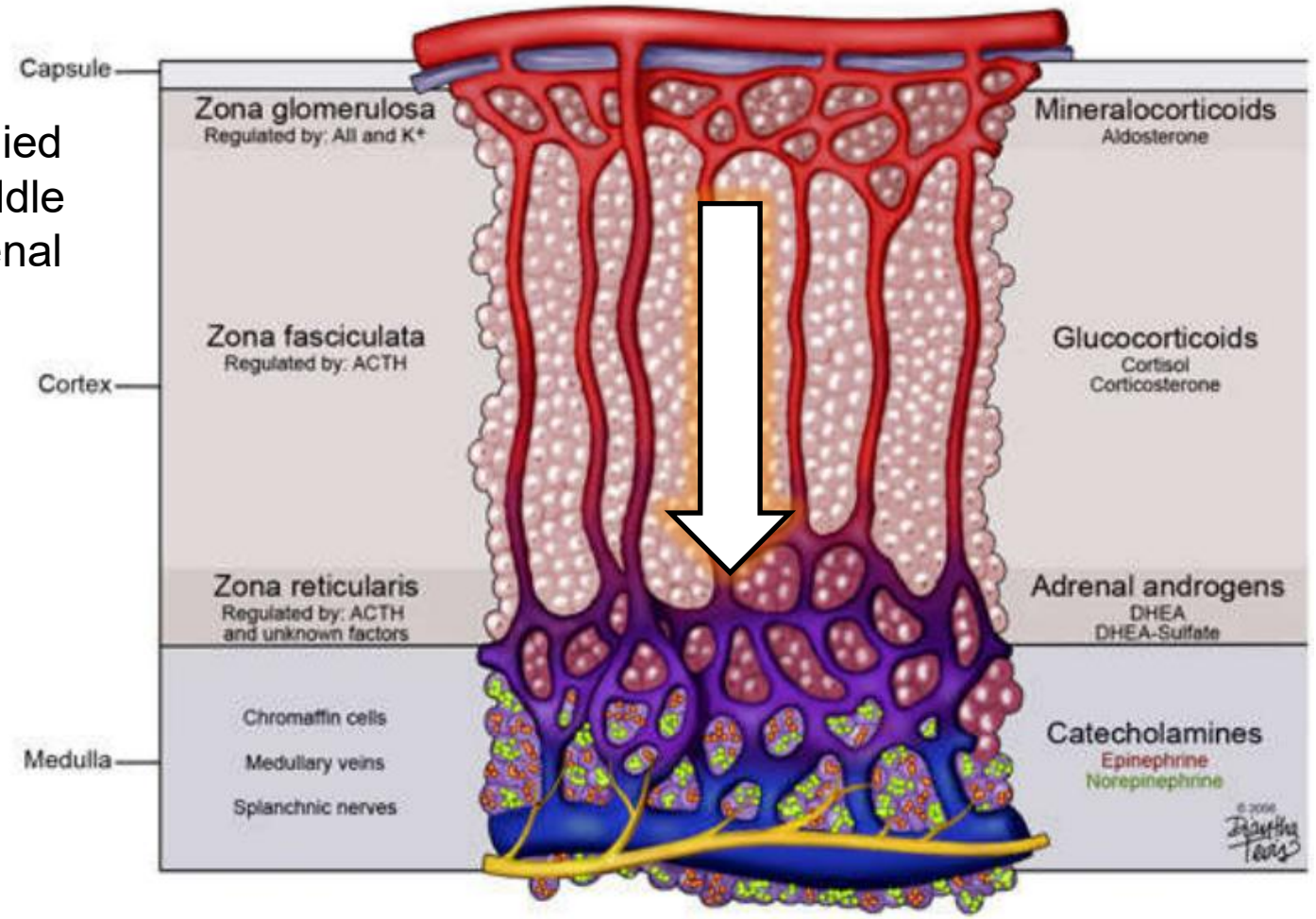
Describe the chemical nature of catecholamines, their biosynthesis, and stimulus for secretion.

Describe the influence of cortisol on the adrenal medulla and explain its physiological significance during stress.



Blood supply of the adrenal gland

Each gland is supplied by the superior, middle and inferior suprarenal arteries



The medullary veins emerge from the hilum of each gland before forming the suprarenal veins

Describe the influence of cortisol on the adrenal medulla and explain its physiological significance during stress.

A patient with primary adrenal insufficiency will also have reduced amounts of which of the following neurotransmitters?

A. Acetylcholine
B. Dopamine
C. Epinephrine
D. Norepinephrine

A patient with primary adrenal insufficiency will also have reduced amounts of which of the following neurotransmitters?

A. Acetylcholine

B. Dopamine

C. Epinephrine

D. Norepinephrine

Synthesis of epinephrine depends on the expression of PNMT, which is under the influence of cortisol. In the absence of cortisol, there will be lower PNMT expression and therefore epinephrine synthesis. Dopamine and norepinephrine are precursors in the synthetic pathway and will be unaffected reduced PNMT. Acetylcholine synthesis is not under the control of cortisol.

Physiological significance of the influence of cortisol on the adrenal medulla

- Cortisol is released during neurogenic and physiological stress
- Cortisol stimulates the production of epinephrine, which is part of the fight or flight response to stress
- Coordinated stress response (cortisol + epinephrine)

Describe the influence of cortisol on the adrenal medulla and explain its physiological significance during stress.

Summary of adrenal medullary function

1. The adrenal medulla synthesizes the catecholamines epinephrine and norepinephrine in a 4:1 ratio, as well as trace amounts of dopamine.
2. Virtually all circulating epinephrine originates from the adrenal medulla. Norepinephrine is also produced in post-ganglionic sympathetic nerve terminals around the body. Catecholamines are carried freely dissolved in the plasma.
3. Epinephrine is produced by the enzymatic processing of tyrosine in the adrenal medulla. Cortisol increases the expression of PNMT, which is the final step in epinephrine production.
4. The blood supply of the adrenal gland facilitates the action of cortisol on epinephrine synthesis, as blood flows from the cortex into the medulla, picking up cortisol as it flows through the zona fasciculata.
5. Cortisol stimulation of epinephrine synthesis enables a coordinated response to stress.

Overall Summary

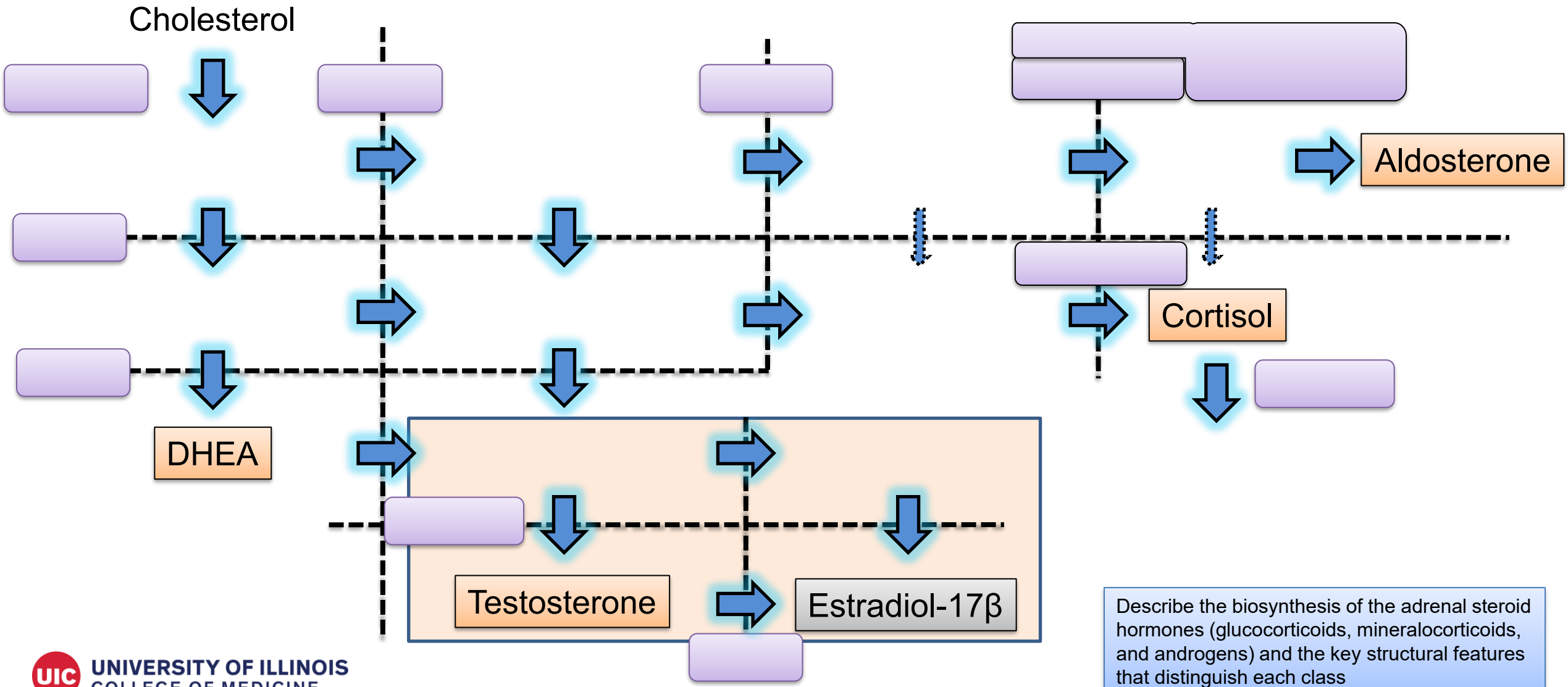
1. The adrenal cortex synthesizes the Δ4 compounds aldosterone (zona glomerulosa) and cortisol (zona fasciculata) and the Δ5 compound DHEA (zona reticularis) from cholesterol.
2. Δ4 compound synthesis requires the activity of 3β-HSD. This enzyme is absent from the zona reticularis. The zona glomerulosa does not express 17α-hydroxylase.
3. Aldosterone and cortisol synthesis involves 21α-hydroxylase and 11β-hydroxylase reactions. Aldosterone synthase performs the 11β-hydroxylase reaction in the zona glomerulosa.
4. DHEA production is dependent on the activity of one enzyme that has two separate enzymatic reactions (17α-hydroxylase and 17,20 lyase). Production is increased by ACTH.
5. Acute stimulation of steroidogenesis involves hydrolysis of cholesterol esters and increased shuttling to the mitochondria via StAR protein. Chronic stimulation involves upregulating activities of P450scc and other downstream steroidogenic enzymes.
6. Cortisol has mineralocorticoid activity that can be prevented by converting it into inactive cortisone by 11β-HSD2.
7. Congenital adrenal hyperplasia can result from deficiencies in enzymes involved in cortisol biosynthesis. The major form is 21α-hydroxylase deficiency. This leads to increased ACTH levels that causes tropic effects on the adrenal cortex and increases production of adrenal androgens in the case of 21α-hydroxylase deficiency and 11β-hydroxylase deficiency. 21α-hydroxylase deficiency causes salt-wasting as it reduces aldosterone production. 11β-hydroxylase deficiency does not cause salt-wasting as the compound 11-DOC has mineralocorticoid activity. 17α-hydroxylase deficiency is another form of CAH that leads to absence of adrenal androgens.
8. Epinephrine is produced by the enzymatic processing of tyrosine in the adrenal medulla and is stimulated by ACTH. Cortisol increases the activity of PNMT, which is the final step in epinephrine production.

Resources to help you master this material

1. Basic comprehension practice questions in Blackboard

2. Blank cheat sheet on steroid synthetic pathways on Blackboard for you to complete

Retrieval practice: Steroid synthesis



Describe the biosynthesis of the adrenal steroid hormones (glucocorticoids, mineralocorticoids, and androgens) and the key structural features that distinguish each class